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Effects of biofuel on engines performance and emission characteristics: A review

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ABSTRACT

Alternative fuels are still needed to compensate for the energy shortages caused by fossil fuel depletion. The paper aims to brief the types of alternative fuels used for the past 30 years. Moreover, it includes the recent types of biofuels (especially biodiesel) and their blends with studies on the performances and the exhaust emissions for different engines. In this study, previous studies were analysed, the challenges faced by the researchers were examined, and incentives for using biodiesel fuel in engines were discussed. The engine performance and emissions when using biodiesels and their blends in different engine models were also surveyed. All biodiesels and their blends have demonstrated the ability to reduce emissions such as carbon oxide (CO), carbon dioxide (CO_2), nitrogen oxide (NO_x), particulate matter (PM) and hydrocarbon (HC) under various operating conditions, as well as the ability to improve the performance of the gas turbine. It is necessary to understand the combustion properties of fuels for their use in an engine. The contribution of this review is to help the engine manufacturers and researchers develop further research relating to readjusting and optimising the biodiesel engine and its relevant system.

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1. Introduction

The need for a pollution-free environment and securing sustainability caused increased global demand for renewable fuels in gas turbines [1]. The aviation industry started back in the 1960s, which is the beginning of gas turbine technology. Rapid progress in technology caused it to become a vital power generation system. The driving force for the improvement of renewable fuels is the increasing demand for sustainable and clean alternative energy sources and global efforts to meet emissions targets by using alternative non-fossil fuels in fuel-flexible gas turbine technologies. Under favourable operating conditions, carbon monoxide (CO), nitrogen oxide (NO_x), and soot emissions can be reduced and regulated using all of the alternative liquid biofuels. The utility of hydrogenated vegetable oil (HVO) and Fischer-Tropsch (FT) fuels is limited due to their high creation cost, confining them to massive scale usage. However, their performance is similar to using jet fuel in aviation gas turbines [2].

The alternatives with the highest prospect for fossil fuel are liquid biofuels with comparable combustion properties and heating value. High moisture, ash, and low energy density are the main difficulties associated with solid biomass, which are avoided by using liquid biofuels, so there is a trend of converting towards liquid biofuels from biomass and biofuel production [3]. The demand for energy by the world has to be fulfilled, which can be done by finding an alternative fuel. The best accessible source to satisfy the energy demand is biodiesel. The prospect of renewable fuels for diesel engines is amongst more than 350 oil-bearing yields specified. Utilising various oils and their mixtures with diesel, tests have been conducted by researchers and scientists [1]. The biofuel

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Abbreviations

SEC	brake specific energy consumption
BP	brake power
BSFC	brake specific fuel consumption
BTE	brake thermal efficiency
CHP	combined heat and power
CO	carbon monoxide
CO_2	carbon dioxide
CPO	crude palm oil
CSOME	cotton seed oil methyl ester
DF	diesel fuel
DI	direct injection
DME	dimethyl ether
EFGT	externally fire gas turbine
EGR	exhaust gas recirculation
ET	engine torque
FT	Fischer-Tropsch
HC	hydrocarbons
HOME	Honge oil methyl ester
HVO	hydrogenated vegetable oil
B20	20% biodiesel with 80% diesel
PB20	20% palm biodiesel with 80% diesel
B50	50/50 biodiesel with diesel
B80	80% biodiesel with 20% diesel
IDI	indirect injection
JB	Jatropha biodiesel
JOME	Jatropha oil methyl ester
Mg	Magnesium additive
MGT	micro-gas turbine
Mo	Molybdenum
MOEE	Mahua oil ethyl ester
MOME	Mahua oil methyl ester
NO_x	Nitrogen oxides
PB	Palm biodiesel
PM	particulate matter
PoB	Polanga biodiesel
SOME	seed oil methyl ester
SVO	straight vegetable oil
THC	total unburned hydrocarbons
TSOME	Tobacco seed oil methyl ester
ULS	ultra-low sulphur
VOME	vegetable oil methyl ester
WPOME	waste palm oil methyl ester
BM	biodiesel-methanol blend
EJB	esterified jatropha biodiesel
B100	100% biodiesel

blending ratio with the traditional jet fuel relies on the feedstock, such as 30% of biofuel blending ratio has used in United Airlines [4].

Compared to petrodiesel, biodiesel has similar combustion characteristics. Because of that, it is the most studied biofuel in gas turbines, especially in small-scale micro gas turbine (MGT) engines. The aims are to enhance combustion performance, reduce pollutant emissions, and increase the stability of lower-grade fuels by developing new combustion technology. High-temperature air combustion is one of the latest techniques in gas turbine combustors. Reduced emissions for CO and NO_x can be significantly achieved in the gas turbine by elevating the air temperature to above 1000 °C at the chamber inlet. Thus, liquid biofuels have low emissions and stable combustion with lower heating values. Catalytic combustion and modest or intensive low oxygen dilution combustion is used as the other extensive technique. When tested in gas turbines combustors, reduction in the emissions of CO and NO_x have been shown by using these technologies. The ultra-low thermal intensity can be achieved by using another technology in combustion, which is the colourless distributed combustion [3].

This paper presents the findings of reviewed critical studies in tabular form, which will facilitate scientists in biodiesel research to get the brisk datum on emission characteristics of exhaust and engine performance for various biofuels and their blends. It also discusses the engine performance and emissions of biofuel (biodiesel, bioethanol, and bio-oil). Moreover, limitations and problems faced by the researchers are also highlighted. Numerous studies have investigated different engine models with different feedstock of biofuels. However, it was found that earlier researchers still could not find a complete replacement for fossil fuel due to the drawback of using biofuels. Based on this fact, the current study tries to address the following research questions:

i. What are the drawbacks that researchers have faced of using biofuel in the engines?
ii. What is the enhancement in running the engines with different biofuels' feedstock?
iii. What are the types of biofuels and kind of engines have been covered until now?
iv. What are the points that previous studies did not cover?

The review method which was employed in the study is shown in Fig. 1 below:

2. Motivation of using biofuels

The most attractive fuel that can replace diesel fuel in diesel fuel engines is biodiesel, non-toxic and biodegradable in nature [5,6]. Biodiesel is one of the renewable sources from an alternative energy source, which the global oil industry has prompted to look at due to the increasing demanding for fossil fuel, causing an increase in its price. Biodiesel gives almost the same functional properties as fossil fuels, and its characteristics are friendly to the environment. For this reason, it is considered the better option [7]. Apart from the sustainable production of biodiesel, its advantages include a great extent of carbon emission reduction, more job opportunities that pertain to the environment, diminishing the need for import fossil fuel and reducing fuel costs by increasing the supply of vegetable oil. With its reserve feedstock, a balance for technology, and current infrastructure availability compared to other countries, Malaysia is ahead in developing palm biodiesel technology [7].

Biodiesel can be used directly in diesel engines without modification. Being a cleaner alternative fuel, it has gained much popularity. In recent years, most researchers and scientists have been using non-edible and edible feedstocks to produce more cost-effective biodiesel blends and enhance the blends' physicochemical properties [8]. For the past few years, biodiesel production has been investigated extensively using different crops of non-edible oilseed [5]. One of the world's leading palm oil producers and exporters, Malaysia opts to use palm oil feedstock because it is economically viable. One of the most efficient oil-bearing crops is palm oil in terms of efficiency, productivity, and land utilisation. On the other side, producing biodiesel from edible oil is not ideal because of its competition as a source for food [9], which has been affected by

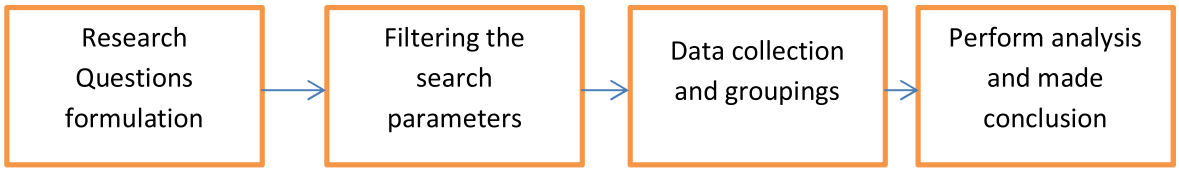


Fig. 1. Flow method of analysis used in this study.

food security [10]. Calophyllum inophyllum and Jatropha are non-edible oils that are planted in subtropical climates and tropical countries. Compared with other non-edible oil sources, Jatropha has an extra advantage; it can survive in abandoned agricultural land because it is a drought-resistant plant. Aside from that, biodiesel fuel could also be transesterified to Calophyllum inophyllum oil. However, Malaysia is still in the nascent state of using Calophyllum inophyllum as biodiesel compared to the jatropha and palm oil industry [9].

According to many researches, when substituting biodiesel in petroleum diesel, the measurement methodology, low-temperature engine performance, modern instrumentation, and the engine itself needs to be modified and optimised, according to many researches [11]. For example, one can improve the economy of the biodiesel engine by using a low heat release engine or a turbocharged engine. Although the influences of engine operation conditions (injection pressure, injection time, load, speed, *etc.*) are not essential to get engine improvement and its control system to optimal match, there is a need for further study to execute those conditions [11]. There is increasing interest when substituting diesel fuel with biodiesel, which causes air toxics from non-regulated emissions regarding diesel. Biodiesel reduces the poly-aromatic compounds and aromatic emissions shown in most of the research based on the engine's operation conditions such as cycle mode, load, etc. [11].

It is an alternative source being explored in most countries worldwide because it is renewable and environmentally friendly. For these reasons, the focus is on substituting petroleum diesel with biodiesel as a part replacement component. Moreover, the choice of alternative fuel with a very high score is biodiesel almost has no sulphur, and its benefits reduce the reliance on fossil fuels. When fuelling the internal combustion engine using biodiesel, the combustion profile is improved due to a higher cetane than petrol-diesel. The well-established advantage of using biodiesel is decreasing the pollutant component of the exhaust. In the last few years, the world witnessed a double-digit annual growth rate of biodiesel production capacities due to the development of 100% biodiesel works [12].

3. Application of biofuel in gas turbines

Versatile combustor, clean combustion, robust mechanical design, and moderate compression ratios are features of a gas turbine, with its steady flame due to the continuous flow of the combustion process [13]. One of the primary sources of uncontaminated power generation is the high-efficiency design of the gas turbine by producing combined heat and power (CHP) [14]. Current gas turbine systems provide a resilient multi-type-fuel supply system with a large power range. They are usually developed based on a single specific fuel such as diesel fuel or natural gas, although they can take different fuels. Currently, the dominating turbine in the global market annually, in terms of the variety of units produced, is the gas turbine, followed by the technology of the steam turbine for generating conventional electrical power [3]. One of the technology approaches is enhancing engine efficiency and saving fuel by adding the recuperator equipped to the cycle. It works on recovering the heat from the exhaust gas to preheat the air through the compressor, as shown in Fig. 2. Another technology used to enhance the output power is Turbine Inlet Cooling System (TIAC). It works on cooling the inlet air to the compressor, as illustrated in Fig. 3.

The overall efficiency is higher compared with the under-firing cooling turbine. This results from the usage of blades in the air cooling turbine from the compressor bleeding, as seen in the study on low heating value fuels fuelled in a combined cycle. Many concerns of liquid biofuels have been addressed in the literature, such as emissions from pollutants, stability of combustion, hazards of flashback, and fuel properties. However, the primary goal of using it in a gas turbine is to reduce the carbon footprint. Different liquid biofuels have been investigated for their characteristics of combustion and spray on MGT and gas turbines in many experiments. There are low emissions of the pollutant with combustion stability provided by the dimethyl ether (DME) fuel [17–19]. The most attractive alternative fuel is biodiesel fuel due to the fossil fuel diminishing supply and its environment polluting nature. CO, carbon dioxide (CO_2), hydrocarbon (HC), smoke opacity, and particulate matter emissions can be reduced by blending biodiesel with diesel due to the similar combustion characteristics of biodiesel and diesel. However, this blend increases the NO_x emission slightly [9].

The appeal in substituting the diesel fuel with biodiesel is due to its ability to use a variety of sustainable and renewable feedstocks, edible oil ranges such as corn oil, soybean, and sunflower, and also non-edibles such as Calophyllum inophyllum, Jatropha curcas (J. curcas), and Moringa oleifera (M. oleifera). Besides, waste cooking oils, animal fats, macro-algae, and microalgae (algae) can also be used to produce biodiesel. Diesel engines can use biodiesel without modifications. Enhancing the engine's performance, reduce diesel engines emissions, and improve fuel physicochemical properties, biodiesel is blended with other fuels [20]. The optimal fuel inlet temperature for jatropha performance was evaluated to be 80 °C instead of 100 °C because of a leakage of lube oil, with consideration for the engine safety, brake specific energy consumption (BSEC), brake thermal efficiency (BTE), and gaseous emissions. A proper substitute fuel for a diesel engine is jatropha oil using a preheated heat exchanger [9]. Compared to Camelina biofuel, the pure jatropha biofuel gave better thrust and lower SFC and fuel flow due to the low heating value. On the other hand, the lowest NOx emissions were for 50% jatropha biofuel blended with jet fuel at all flight conditions [21].

3.1. Biodiesel

Over the years, many studies about biodiesels and their blends have been carried out, with many papers published each year. Various aspects of biodiesel research have been published in many review papers in the scientific literature. The most studied biofuel in a gas turbine is biodiesel, specifically the small-scale micro-turbine [8]. Biodiesel allows the usage of current gas turbines of land-based industries, fuel-robust with slight modification. However, it needs strict obedience to the characterisation of jet gas

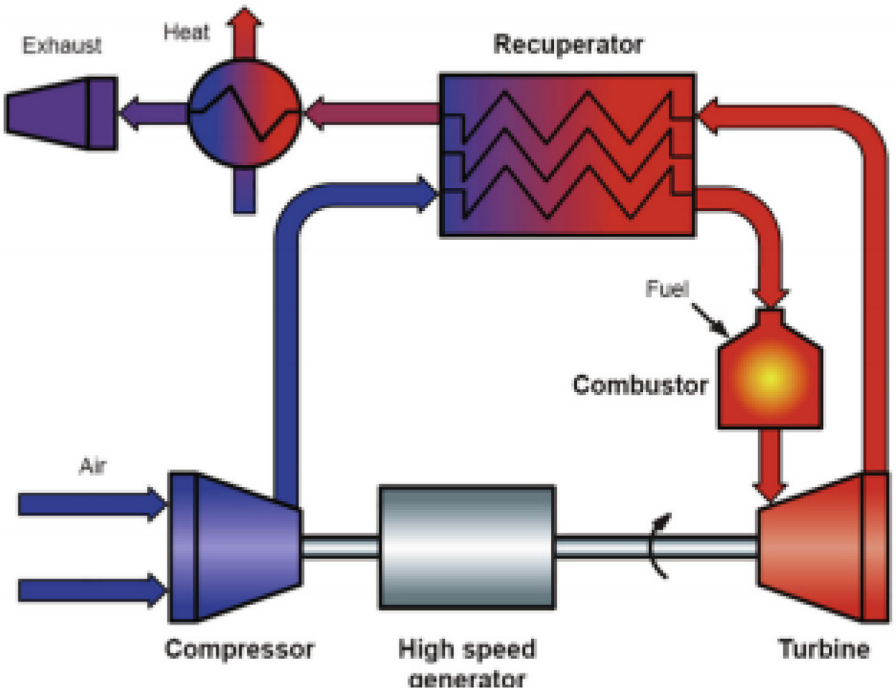


Fig. 2. Schematic of combined heat and power system (CHP) in micro gas turbine cycle [15].

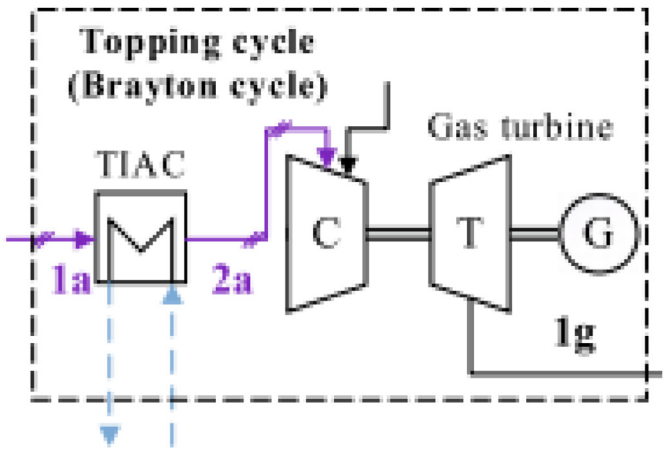


Fig. 3. Schematic diagram of using turbine inlet cooling system (TIAC) into gas turbine cycle [16].

description biodiesel spray flame using the kind of a gas turbine burner in an aviation-based gas turbine application. The potential of gas turbine fuels as biodiesel has been investigated by using swirl flame burner type into the gas turbine scale.

The benefit of using a simplified burner permits the parametric studies containing combustion performance comparison, baseline fuels with emissions under a well-controlled environment—biodiesel application in a gas turbine for electrical energy, and the structure of the flame. Using an actual gas turbine to test biodiesel combustion performance has been investigated, containing aviation, MGTs, and industrial gas turbine engines. Off-grid power generation for rural regions, households, and small businesses are some applications of MGT. Lower soot, NO_x, and CO emissions have been achieved by using biodiesel. It prohibits the formation of soot and assists local combustion by the content of fuel-bound oxygen. Compared to jet fuels, the requirement of using biodiesel in a gas turbine is less stringent, making biodiesel suitable for stationary gas turbines [2]. When biodiesel contains unsaturated compounds, there is a more significant influence on the emission of NO_x. However, when the content of unsaturated compounds is more extensive, the NO_x emission will be reduced [22].

Biodiesel's combustion characteristics were similar to diesel and were observed to perform better than enzymes and acid catalysts for the base catalyst. It was also inferred that using vegetable oil/diesel blend produced an inferior engine performance as the high viscosity oil contaminated the lubricating oil and injector coking. A considerable performance in improvement was observed through blends of refined oil. Compared with diesel, the emission of NO_x and unburnt hydrocarbon (HC) from the engine was found to be higher in all fuel blends [1].

Exhaust emission characteristics and engine performance of diesel engines are the aspects of recent reviews. These reviews also pertain to the effects of biodiesel and its blends on the characteristics of single exhaust emission, particulate matter (PM), and biodiesels produced from a specific group of biodiesel blends such as biodiesel-alcohol blends or single feedstock are fuelled into the diesel engine. Recognising the papers about this subject matter through the abundant literature can be daunting due to the number of documents related to biodiesels and their blends is ever-increasing each year [8]. One of the approaches to improve exhaust emission characteristics and the engine performance of diesel engines and lessen the reliance on fossil fuels and edible oils for producing fuel is blending different types of biodiesel [23,24]. Engine performance will be enhanced significantly by blends with high capability to form combustible mixtures with air, such as high flashpoints. Over the years, bio-oils have fuelled internal combustion engines, and the engine performance and exhaust emissions have been investigated [8].

Commonly, engine power reduction can be accepted when using biodiesel. Also, when using biodiesel, it is not easy for drivers to perceive power losses during partial load of practical driving, particularly for blended fuel with a small portion of biodiesel. The

comprehensive viewpoint agreed that the reduced heating value of biodiesel is the main reason for power loss compared to diesel. There is no unanimous conclusion that there are specific effects on engine power caused by the high lubricity and high viscosity of biodiesel. Also, there is no essential effect on engine power from using biodiesel feedstock. An additive is advantageous to recover power for the biodiesel engine, improving biodiesel combustion and ignition [11].

The combustion performance of the diesel engine may be further improved by using additives to improve the biodiesel properties [11,25]. This will improve engine power and, at the same time, improve efficiency. A metal-based additive has less efficiency in improving hydrocarbon (HC) emissions than other emissions for biodiesel. Also, a small proportion of methanol and ethanol added into the biodiesel blend may be advantageous for HC emissions [11]. Oxidised biodiesel impact on the engine performance and emissions have been evaluated. Compared to other engines having diesel fuel, the engine performance was similar for neat biodiesel and their blends with the same thermal efficiency. However, the fuel consumption was higher compared to diesel. Oxidised neat biodiesel produced 16% lower hydrocarbons and 15% lower exhaust carbon monoxide than unoxidised biodiesel [12].

3.2. Bioethanol

Bioethanol is flammable, low toxicity, colourless, and biodegradable liquid. It has a −78 °C pour point and a low viscosity of 1.5 mm^2/s, comparable to jet fuel. Bioethanol commonly replaces gasoline to power the internal combustion engine. E85 (85% biodiesel/ethanol with 15% gasoline) is often used for bioethanol blends. There is a significant difference between bioethanol and diesel properties as the former has high vapour pressure, low viscosity, and high flashpoint. Storing this fuel and delivering the fuel to the gas turbine requires some modification. Although bioethanol is applied widely in a reciprocating gasoline engine, studies are relatively scarce for bioethanol in a gas turbine [2].

3.3. Bio-oil

The viscosity of straight vegetable oil (SVO) is higher than conventional fossil diesel when applied in a gas turbine system. This poses a technical challenge as the highly viscous oil will negatively impact fuel flow delivery, leading to an inferior spray atomisation process. Using a twin-fluid injector and adding duel preheating capability to modify the fuel delivery system allow various control parameters. Another strategy to maintain the low viscosity is by blending SVO with conventional fuel [2].

Table 1 shows the engine fuelled from biodiesel fuel and its blends. Most studies have demonstrated that the diesel engine can be used without modification by using biodiesel, leading to enhancement of performance and emission characteristics. However, raw biodiesel fuels cannot be used as a fuel for the engine, so it needs to be modified or blended with petroleum or another type of biodiesel.

Table 2 displayed the results of the brake thermal efficiency (BTE), brake specific fuel consumption (BSFC), ignition delay and power output of using jatropha oil methyl ester (JOME) and Calophyllum inophyllum biodiesel. The studies found that using JOME and Calophyllum inophyllum increased the brake thermal efficiency, but one study reported that the BTE decreased when JOME was used. Also, JOME showed improved BSFC, but Calophyllum inophyllum's BSFC decreased. The power increased at high speeds when JOME was used, but there was no significant effect on power when Calophyllum inophyllum was used.

Table 3 shows the different engine performances between biodiesel and diesel. M. olefiera led to increasing brake power by up to 1% compared to J. curcas as it has a higher calorific value. The brake specific fuel consumption increased by 10%–14.5% when using palm oil compared with diesel. This is because palm oil has a lower calorific value compared with diesel situation.

Increasing the proportion of biodiesel in the mixture will adversely affect the BSFC. Raheman [33] found that an increase in the blend ratio of the mahua oil produced led to an increase in the BSFC, Altiparmak [42] noted that the level of specific fuel consumption (SFC) increased at low speeds, depending on the amount of the tall methyl ester oil. The conclusion from other studies is that due to reduced fuel viscosity, the BSFC decreased, for example, by Parmanik [38]. While another study found that when the oxygen content of the fuel increases, the BSFC will decreases [27]. As for operating conditions such as engine speed, a study found that SFC is affected at high engine speeds [45], while another found that SFC is not affected at high speeds [42]. Thus, SFC depends both on the type of biodiesel and the engine speed.

The biodiesel properties influence the brake specific fuel consumption and the brake power. Among the most critical factors that affect fuel consumption is the calorific value. For example, Sharon et al. [54] found that BSFC increased by 10% for palm oil, while Tan [55] found that BSFC increased by 4.77% compared to diesel due to decreased calorific value palm oil fuel. Experiments have proven that BTE increases over the use of biodiesel compared to diesel. Puhan [44], for example, found that BTE increased by 25.42% when using mahua oil methyl ester (MOME).

In comparison, 50% of the jatropha and diesel fuel mixes obtained an acceptable BTE for the diesel engine, but Ramadhas [39] found that the rubber seed oil blends with diesel had lower BTE than diesel. The use of different types of biodiesel and its blends have improved the performance of the engine. Examples include Stayer [30] who enhanced engine performances when running with high erucic rapeseed oil and degummed canola oil. Kinney [43] who used soybean oil, McDonnell [35] who found that the 25/74 mix drew the best performance from the engine; and Al-Widyan [36] who obtained the best performance for a 75/25 blend of waste cooking oil/diesel.

4. Challenges and limitations

4.1. Concerns on using biodiesel

Several challenges and issues face biodiesel production. These include limited land for deforestation and usage, food versus fuel prioritization, feedstock issues, sustainability, and tough global competition [7]. There are no recent reviews on the engine performance and exhaust emission characteristics of a fuelled diesel engine with various biodiesel blends and recent development of technologies to reduce emissions [8]. Of the 27 kinds of literature reviewed, 70.4% studied the effect of pure biodiesel on engine power. They agreed that the engine power would drop with biodiesel, specifically with pure biodiesel, due to the loss of biodiesel heating value. However, the power loss found by some authors was lower than expected compared to diesel due to power recovery [11].

The low heating value for biodiesel is the main reason that causes an increase in fuel consumption, along with high viscosity and high density [11,25]. The different production processes and qualities impact engine economy or different carbon chain lengths and heating values with different biodiesel feedstocks [11]. Testing the durability of biodiesel engines requires spending extra time and costs more than those tests in emissions, engine, and economy. Because of this, only a small number of scientists are devoted to it. The following aspects for durability studies were focused on: fuel system problems, carbon deposit, and engine wear [11].

Table 1
Biodiesel performance.

Ref.	Fuel	Type of engine	Performance
[26]	Six variants of sunflower oil	Diesel engine	Prevented engine failure by dewaxed and degummed vegetable oil. The diesel engine can use modified sunflower oil as a fuel, but the raw sunflower cannot be fuel.
[27]	Vegetable oils such as soybean oil, peanut oil, sunflower oil, and cottonseed oil	Direct injection (DI) and indirect-injection (IDI) diesel engine	There were positive results for the IDI engine when oxygen is present because BSEC decreased as the oxygen increased in the fuel content, but it is inversely affected with nitrogen. It was found that the fuel difference is more acceptable for the IDI engine than the DI engine.
[28]	Sunflower, safflower and rapeseed oils	Single cylinder diesel engine	Equivalent to output engine power by using diesel fuel. Carbonisation causes severe problems for long-term durability.
[29]	Blends of rapeseed oil	Indirect injection diesel engine	Increasing rapeseed oil proportion in the blend causes an increase in the power output, so neat rapeseed oil produced less power output.
[30]	High erucic rapeseed oil and degummed canola oil	Large and small diesel engine	Increased specific fuel consumption (SFC) and particulate matter for these fuels. Operating the engine with degummed canola oil produced better engine performance than the engine with crude canola oil.
[31]	Blends of palm oil with diesel	Ricardo E−6 and Lister diesel engines	There was no sign of adverse combustion chamber wear or lubricating oil or increased carbon deposit using palm oil in a short-term test.
[32]	Blends of sunflower oil with diesel	33 kW DI diesel engine with three cylinders	–
[33]	Blends of mahua oil and diesel with high-speed diesel	E6 Ricardo engine	It was found that the compression ratio and injection timing test are inversely proportional to the engine load. With the increase in the percentage of biodiesel in mixtures, the brake thermal efficiency (BTE) decreased, and brake specific fuel consumption (BSFC) increased.
[34]	Mustard oil	Direct injection engine with 4-cylinder and turbocharged	A power engine is developed to be equivalent to diesel power. Tests have to be carried out for the long term.
[35]	Semi-refined rapeseed oil	–	25/74 blend for rapeseed oil/diesel had the best performance for the engine. Although there was no contamination of lubrication oil or wear on engine components, the carbon build-up shortens the injector life.
[36]	Blend of waste vegetable oil ethyl ester with diesel (75/25, 50/50, 25/75)	Direct injection diesel engine	These blends had a better fuel economy. Waste cooking oil/diesel blend (75/25) gave the best performance.
[37]	Crude palm oil	Diesel engine	Observed piston wear rings, delivery valve and plunger of the injection pump, cylinder lining scuffling, formation of uneven sprays and combustion chamber had heavy carbon deposits.
[38]	Jatropha oil and diesel blends	Diesel engine	When compared to neat vegetable oil, the engine performance improved significantly. The viscosity decreased for vegetable oil, the specific fuel consumption (SFC) was also reduced. 50% jatropha oil had obtained acceptable thermal efficiencies for the engine.
[39]	Rubber seed oil blended with diesel (20/80, 40/60, 60/40, 80/20)	5.5 kW direct injection diesel engine with a single cylinder	The blends developed heavy carbon deposits, even though acceptable specific fuel consumption (SFC) and thermal efficiency had been obtained with 80/20 blends.
[40]	Blends of rubber seed oil with diesel	Diesel engine with a single cylinder	Compared to diesel, biodiesel blends had lower BTE. Biodiesel recorded higher combustion in lower duration heat.
[41]	Waste frying oil	Direct injection diesel engine with a single cylinder	Compared to diesel, it consumed marginally higher fuel. It was observed that the use of preheated waste frying oil improved the engine's performance and reduced the emissions of CO and smoke, so it can be used to operate the engine in the short term.
[42]	Ester methyl of tall oil blended with diesel	Direct injection diesel engine	Engine torque increased by 6.1% and power output by 5.9% when using tall oil methyl ester-diesel fuel and its blends at high engine speeds. Depending on the amount of tall oil methyl ester, SFC increased at lower speeds but did not increase significantly at higher engine speeds.
[43]	Soybean oil	Diesel engine	It is used to enhance the engine diesel performance.
[44]	Ethyl ester of mahua oil (madhuca indica oil)	Direct diesel engine with four strokes, naturally aspirated	The results showed that mahua ethyl ester oil could be used as an alternative fuel instead of diesel. The BTE of mahua oil was 26.42%, while diesel was 26.36%.
[45]	Methyl ester neat rapeseed blend with diesel (5%, 10%, 20%, 35%)	Unmodified direction injection diesel engine, naturally aspirated, four cylinders and four strokes	The inclusion of methyl ester rapeseed in diesel fuel was examined to affect BSFC for a high-speed diesel engine.
[46]	Blends of esterified Jatropha biodiesel (EJB) are (4%, 10%, and 20%)	Low heat rejection engine & conventional engine	The optimum blend was B10. It reduced the BTE value and the emissions (NOx, CO, HC) for both engines

Table 2
Effect of biodiesel feedstocks on performance parameters.

Fuel	Brake thermal efficiency (BTE)	Brake specific fuel consumption (BSFC)	Power
Jatropha biodiesel (JOME)	• 29% vs diesel [47] • 22.44% at 50% jatropha oil (JO) [38] • Improves by 8.9% at 250 bar injection pressure and compression ratio (CR) 18 [48] • Max 27.4% JO, 29% JOME, 28.1% JO methanol [49], • 4.9% by using exhaust gas recirculation (EGR) 0 range 0−25% [50]	Improve by 10% at 250 bar and CR 18 [48]	+3.8% using jatropha biodiesel JB50 at 2000 and 2100 rpm [51]
Calophyllum inophyllum biodiesel	Higher at 100% load [52]	Lower at 100% load [52]	No significance at lower speeds of 1200 and 1400 rpm [51]

Table 3
Comparison between applications of biodiesel and diesel fuel.

Ref.	Type of Fuel	Performance	Justifications
[53]	M. olefiera vs. J. curcas biodiesel	1% brake power (BP) for M. olefiera	Because its calorific value is higher than J.curcas biodiesel.
[54]	Palm oil vs. diesel	BSFC is 10% higher for palm oil	Because palm oil has a lower calorific value by 14.4% vs. diesel.
[54]	Palm methyl ester (PME) vs. diesel	2.6–14.5% BSFC for PME	Undesirable to increase BSFC because higher BSFC means more fuel is consumed and BP is increased.
[54]	PME vs. diesel	0.3–2.2% BTE for diesel	Desirable for BTE to be higher because that indicates the crankshaft is driven in the diesel engine using the mechanical energy converted from the energy of the burned fuel.
[55]	Palm oil biodiesel vs. diesel	4.77% specific fuel consumption (SFC) increased for pure biodiesel	Viscosity decreases, BSFC increases. The calorific value for biodiesel is lower than diesel, which means biodiesel fuels have lower energy content.
[56]	Palm methyl ester vs. diesel	Flashpoint (FP) 130.7% for PME	FP reduces the risk of fire hazards.
[57]	PB20 (20% palm biodiesel and 80% diesel) vs. DF (diesel fuel)	3.88% BSFC increase for PB20	Incomplete fuel combustion is due to the high kinematic viscosity. Because PB20 contains oxygen, it caused a lower heating value for PB20.
[46]	B04 (4% of EJB), B10 (10% of EJB), and B20 (20% of EJB)	The increasing of BSFC value with increasing the blending volume	That because biodiesel has a lower calorific value and higher density compared with diesel

Even though particulate matter (PM) emissions for biodiesel remain very low compared to diesel, PM emissions might deteriorate by using EGR. In the case of low-temperature conditions, in comparison with diesel, PM emissions will increase abnormally for biodiesel, so it is worth investigating this issue further. On the other side, NO_x emissions will be reduced by using EGR. However, the EGR rates might not match well with the same conditions of biodiesel engines due to the trade-off of biodiesel combustion characteristics. After all, it is optimised to match the operational requirements of diesel. This area of research needs to be further explored and refined [11]. PM emissions value can be improved by adding oxygenates to biodiesel, but it would not be valuable for power recuperation. Metal-based additives may be effective because it reduces biodiesel PM emission due to the catalyst effect. In the future, comprehensive assessments on engine performances and other emissions are required by using emulsifiers, metallic additives, oxide additives, etc., due to their effect on improving NO_x emission of biodiesel [11].

Higher cetane numbers will lead to lower premixed combustion and lead to early burning, leading to softer changes in temperature and pressure. That causes the formation of NO [11]. CO emissions have been proven to be significantly influenced by engine load. Most conclusions about CO emissions being affected by engine speed are unanimous, so increasing the engine speed will decrease biodiesel emissions. This points to an essential role for an oxidative catalytic converter (which is designed for diesel engines) with CO emissions for biodiesel. However, it may become weak in terms of its conversion efficiency. The conclusion about the hydrocarbon (HC) emissions for biodiesel being affected by engine load is inconsistent. The function of the oxidative catalytic converter seems to be weakened, although it has a positive impact on HC emissions for biodiesel [11].

The attention of researchers/scientists all over the world has been toward getting additional biodiesel alternative fuel for diesel and still eco-friendly. A renewable fuel containing the alkyl monoesters of fatty acids and from animal or vegetable fats is biodiesel. The oxidation is one of the drawbacks of biodiesel compared to petroleum-based diesel fuel. In its superior levels, the fuel become acidic and shape indissoluble sediments and gums due to oxidation [12].

4.2. Concerns on using bio-oil

The main challenge for the external fire gas turbine (EFGT) technology is that it has an excessive-temperature heat exchanger with a decrease in turbine inlet temperature of 700–900 °C. Although several studies utilised liquid biofuels with other outwardly fired thermal engines (especially in steam boilers), there may be an absence of studies using EFGT with liquid biofuels [3].

Low volatility and high viscosity are the primary drawbacks of vegetable oil that have to be mitigated for any use in diesel engines. *Trans*-esterification, thermal pyrolysis or cracking, dilution of biodiesel fuel, preheating the oil, and micro-emulsion in solvents are ways to reduce viscosity [9].

5. Results and discussions

The biodiesel feedstocks' physicochemical parameters have been measured and reviewed according to the EN and ASTM standards. Most results of analysing biodiesel properties found that they were close to diesel fuel. Even though a few biodiesel samples did not meet the standard, they did not significantly affect the fuel properties performance. The feasible alternative without any modification is biodiesel, so it is the natural substitute for diesel. On the contrary, the volatility is reduced, and viscosity is increased due to the enrichment of the biodiesel [5].

Applying biodiesel in an aviation gas turbine showed possible results with benefits of reducing the emissions but at the expense of performance. The physical properties for Fischer–Tropsch (FT) and HVO are similar to jet fuels. HVO and FT fuels have long-chain hydrocarbons without the oxygen molecule and contain energy density resembling jet fuel. The lower pour point is the main advantage of these fuels that enables them to be applied in an aviation gas turbine (at high altitudes, especially). The tests have shown comparable performance with the aviation gas turbine using jet fuel; meanwhile, the particulate matter emissions improved. The high production cost of using HVO or FT fuel is the main limitation of using them extensively at the present stage. The biodiesel physical properties are slightly weaker than conventional fossil fuel, the density and viscosity are slightly higher, the pour point is high, and the energy content notably is lower [2].

On the other hand, jet engine thrust is improved, and exhaust gas temperature is reduced when alternative jet fuel is used to run the RU19A-300 gas turbine [58]. The effects of combustion products and fuel have been extensively investigated on the path of hot gases and turbines. Furthermore, the techno-economic aspects have considerably influenced the biofuel used in gas turbines, including fuel co-firing, heat recovery, and combined heat and power (CHP) configuration [3].

A vast majority of the scientists reported that the long-term test results showed engine failure due to lubricating oil contamination and higher carbon built up, but short-term engine tests using vegetable oils as fuels were very promising. Blending vegetable oils with diesel altered the fuel chemically, preventing engine failure. It was reported that the biodiesel combustion characteristics resemble that of diesel and have a shorter ignition delay for blends,

peak heat release, and higher ignition pressure and ignition temperature. It was found that output engine power is equivalent to that of diesel fuel. It is also observed that the enzymes and acid catalysts are less effective than base catalysts [1]. The vast majority of literature reported that when using biodiesel, the NO_x emissions would increase, mainly due to the higher oxygen content of biodiesel.

Furthermore, NO_x emissions for biodiesel are also affected by different injection characteristics and cetane numbers [11]. Different types of combustion chamber experiments had been conducted. It was found that a spherical combustion chamber gives better results compared to other types of combustion chambers. Several tests have been done for different processed and raw vegetable oils such as soybean, rapeseed oil, palm oil, and sunflower oil [1].

For biodiesel-diesel blends, increasing the percentage of biodiesel blends lead to an increase of BSFC and BP, while the total unburned hydrocarbons (THC) and BTE tend to decrease. Based on the results available in the existing literature, it is not possible to generalise the trends, and these results should be interpreted with caution. Such as, increasing the percentage of C. inophyllum biodiesel blends lead to an increase in CO emissions but increasing the palm oil biodiesel blends lead to a decrease in CO emissions. Generally, the increase or decrease of exhaust emissions and engine performance parameters depends on the biodiesel composition and biodiesel types and diesel in the blend. For mixed biodiesel-diesel blends, BSFC tends to be higher while BTE, engine torque (ET), and brake power (BP) tend to be lower. However, the differences are within a range of 3–10% and are not significant. For mixed biodiesel-diesel blends, the THC and CO emissions are lower compared with diesel. The characteristic of biodiesel blends is that the NO_x emissions are still higher, with a 5–7% difference for these blends [8].

The most crucial factor leading to reducing the PM emissions of biodiesel is the cetane number for biodiesel. There are other factors, such as lower sulphur and aromatic compounds. PM emissions of biodiesel will be higher due to the bigger engine load; most researchers can accept this. The PM emission will be lower due to the higher engine speed; there is no objection to this trend. The average gas temperature in the combustion chamber will increase due to the increased overall fuel-air ratio as the load is increased. Hence, NO_x formation, which is sensitive to temperature, also increases. The level of NO_x emissions for biodiesel will be higher when the engine load is more significant, which is in line with the mechanism of NO_x formation. When using biodiesel, CO emissions are reduced due to the lower carbon and higher oxygen content to hydrogen ratio in biodiesel compared with diesel; this is commonly accepted. CO emissions decrease with the increase in pure biodiesel content in the fuel blends.

Feedstock and other properties of biodiesel affected the CO emissions in terms of advances in combustion and cetane number. When used pure biodiesel instead of diesel, hydrocarbon (HC) emissions are reduced; this viewpoint is predominant. Increasing biodiesel content reduces HC emissions for biodiesel, which is shown in most of the research. HC emissions are affected by the feedstock of the biodiesel and its properties, such as the different saturation levels or the chain length of biodiesels. Lower HC emissions are preferable for combustion and advanced injection of biodiesel. Some researchers indicated that a low carbon to hydrocarbons ratio caused a reduction of CO_2 emissions; other researchers showed that the more effective combustion leads to increasing or similar CO_2 emissions; these conclusions are not inconsistent.

Nevertheless, the CO_2 life cycle circulation view is that the CO_2 emission of biodiesel is reduced significantly. There are varying results of carbonyl compounds' emissions for biodiesel, but it is widely accepted that these emissions are increased by biodiesel due to the higher oxygen content. Likewise, these emissions are affected by engine types, operation conditions such as cycle mode and load, and methanol increasing results for acetaldehyde emissions [11].

It has been found that high viscosity and density and low heating value of biodiesel lead to increased biodiesel consumption. However, this trend will be weakened as the biodiesel proportion is reduced in the blend. Limited literature concluded that reducing the wear and carbon deposit of the key engine is preferable when using biodiesel, compared to diesel. The soot formation is lower, which inherently reduced the lubricity and particulate matter (PM) emissions of biodiesel [11]. Table 4 shows the main types of alternative fuels, their feedstock and the emission exhaust for specific engines.

Table 5 illustrates the effect of using alternatives fuels in different engines. A study has shown that a higher ignition temperature and pressure due to using jojoba methyl ester causes the ignition delay to be lower. One study reported that combustion efficiency improved when water was emulsified with diesel.

Table 6 described the emissions of engines fuelled from biodiesel fuels and their blend. It shows that the carbon deposit number increased on the injection tip when mixing peanut oil and sunflower with diesel into a single-cylinder engine. Additionally, the nitrogen oxide (NO_x) and smoke reduced slightly, and there was a 50% reduction of HC when the soybean ester was used. In addition, the smoke, NO_x, and carbon oxide (CO) were reduced when a rapeseed methyl ester blend with diesel was used due to the increased inlet air temperature. When tallowate methyl and ethanol blend with diesel was used in a direct injection engine, the carbon dioxide (CO_2) and hydrocarbon (HC) did not change, but CO was reduced. Also, CO, CO_2, NO_x and smoke emissions are decreased when the olive methyl ester was put into a direct diesel engine. Finally, all emissions except carbon monoxide decreased when the mustard oil was preheated at all loads. Adding a tobacco seed oil to diesel caused the reduction of sulphur oxide (SO) and CO emissions but increased the NOx emissions slightly.

Table 7 described the types of biodiesel blends with examples of these types. There are seven ways of blending: i. Blend the feedstock of biodiesel with diesel in different volumes (e.g. 75% of crude palm oil with 25% of diesel), ii. Blend two types of biodiesel with diesel (e.g., 15% coconut, 15% palm oil, 70% diesel), iii. Blend biodiesel-bio-oil with diesel, iv. Blend non-biodegradable material with biodiesel, v. Blend biodiesel with synthetic liquid fuel, vi. Blend biodiesel with alcohol (methanol or ethanol) to reduce the combustion temperature and increase the heat capacity, vii. Use nano-additives with biodiesel.

Table 8 displayed the types of engines running on different blends of biodiesel with diesel. Various blends were used to run different engine models, such as the B10–B100 blend to run the ID engine. Polanga and Karanja oil and their blends with diesel used to test the tracer engine, Nissan diesel engine tested by using waste cooking oil ethyl ester (WCOE), waste cooking oil methyl ester (WCOM) and their blends, and biodiesel-bioethanol diesel and its blend used to operate the standard rail direct diesel engine.

Table 9 shows the test operation hours for the engines with different fuels, such as a three-cylinder DI engine running waste olive oil for 50 h and palm oil with various volumes used in a four-cylinder ID engine for 100 h.

Table 10 shows the effect of the parameters on the CO, NO, and total unburned hydrocarbon (THC) emissions. The oxygen content, volume percentage of biodiesel, and engine load are the most used parameters to enhance the effect of the emissions. CO and THC emissions decrease, but NO_x emission increases while the O

Table 4
Combustion tests with the different methods of atomisation.

Fuel	Engine/burner	Atomisation method	Feedstock	Emissions
SVO	Micro gas turbine (MGT) (DG4M − 1, Capstone 30, Garret GTP30-67) [59–61], Swirl flame burner [62–64], Lean premixing and pre-vaporising burner [65,66]	Pressure swirl [59,61], Flow-blurring [62], Air-assisted [60,63–66]	Soybean [62,64], Sunflower [60,67], Jatropha [63], Rapeseed [65–67]	There are different results based on the operating conditions
Biodiesel	Swirl flame burner [63,64,68–73], Coflow burner [71], MGT (Capstone C30) [74,75], MGT (30 kW) [76,77], semi closed-cycle gas turbine [78], gas turbine engine (GE 6531B) [79], gas turbine sector rig (Allison/Rolls Royce T56-A-15) [80], helicopter turboshaft T63-A-700 [81], turbojet engine (Armfield CM4) [82], turbofan engine (CFM56-7B) [83], gas turbine (IS/60 Rovers) [84]	Pressure swirl [69], Air-assisted [63,64,68,70,72,73], Flow-blurring [85], Pressure [71]	Palm [68,69,72,82], Jatropha [63,84], Canola [70,77,80], Soybean [64,73,75–78,85], Rapeseed [68,71,72,77,79], Chicken fat [64], Fish [80]	There are different results based on the operating conditions

Table 5
Biodiesel combustion applications using different engine models.

Ref.	Fuel used	Engine	Findings
[86]	Soybean methyl esters blend with diesel	Modified injection diesel with a bowl in a piston and swirl type in medium, turbocharged	Combustion characteristics are the same as diesel and shorter ignition delay due to the blend.
[87]	Jojoba methyl ester	Shock tube test rig	Jojoba methyl ester reported higher IT (ignition temperature) and IP (ignition pressure) with lower IDP (ignition delay period).
[88]	Tallowate methyl esters	306 kW diesel engine	Diesel methyl tallowate peak rate of heat release was lower compared to diesel.
[89]	Waste cooking oil	Direct injection diesel engine	There was a general perception that waste cooking oil might harm the fuel injection system due to its high viscosity. However, it was noted that waste cooking oil has less viscosity than diesel in the pre-combustion stage due to the shorter ignition delay.
[90]	Waste vegetable oil ethyl esters	Furnaces and boiler combustors	It was found that biodiesel burned more efficiently than diesel fuel when the lower energy rate was used.
[91]	Rice bran oil	Diesel engine transport	B20 has a lower heat release rate and ignition delay during the premixed combustion phase than mineral diesel.
[92]	Sunflower oil methyl ester	Turbocharged IDI diesel engine	It was found that the maximum power and thermal efficiency was when using 17.5% biodiesel. The exhaust temperature of the mixture is lower than diesel fuel at full load.
[93]	Water-in-diesel emulsions	Diesel engine (DI)	When water is emulsified with diesel, it improved the combustion efficiency.
[94]	Biodiesel	Direct injection Yanmar diesel engines	Whether biodiesel is neat or blends form, the fuel injection and ignition processes are altered.
[95]	50% blend of Papaya seed biodiesel (PSB)	Kirloskar single-cylinder diesel engine	It enhanced the engine performance as decreased the BSFC value and increased the BTE value due to the oxygenated additives

content increases. In addition, most of the emissions are reduced by the biodiesel blends, except the NO_x shows that it depends on the quality of the feedstock.

On the other side, the influence of using biodiesel fuels on PM, NO, CO, and HC emissions are shown in Tables 11–14, respectively. As Table 11 illustrated, the PM emissions decreased when different biodiesel blends were used, such as Karanja, Jatropha, Polanaga, sunflower, and olive oil. Kalligeros [111] found that the PM increased when using pure palm oil.

Most of the studies concluded that NOx increased when biodiesel was used with different conditions, as shown in Table 12. Nabi [176], Lujan [159], Lertsathapornsuk [177], and Hansen [178] found that B100 increased NOx emissions at high load and high speed. There are a few studies such as Puhan [44], Dorado [112], Keskin [158], and McCormick [179] who found that NOx emissions become lower when using biodiesels such as mahua methyl ester, B60 with added magnesium (Mg) and molybdenum (Mo) additive, blends of 20% pure biodiesel with ultra-low-sulphur (ULS) diesel, mahua oil ethyl ester (MOEE) and waste olive oil.

As shown in Table 13, biodiesel decreased the effect of CO emissions such as Karanja methyl ester, waste Palm oil methyl ester (WPOME), seed oil methyl ester (SOME), Honge oil methyl ester (HOME), *etc.*

Many studies found that the hydrocarbon (HC) emission effect was reduced when biodiesel was used, as shown in Table 14. For example, using cooking oil, rapeseed, soybean and diesel in a heavy-duty engine reduced total unburned hydrocarbons (THC) when the biodiesel volume was increased. This is also true for mahua methyl ester (MME), biodiesel-methanol blend (BM), Jatropha biodiesel (JB), and their blends. In contrast, Kumar [193] and Banapurmath [47] are found that HC increased slightly. Such as, HC emission increased when Jatropha methyl ester, Honge oil methyl ester, Jatropha oil methyl ester (JOME), and seed oil methyl ester (SOME) were used when compared with diesel.

Each type of biofuel had different effects on the emission characteristics and engine performance, as shown in Figs. 4 and 5. The lowest value for CO emission was found by using VOME fuel. In comparison, the lowest HC emissions value was achieved by B100 (B100 refers to the study [159]). As well as, the MOEE gave the lowest value for NOx emissions compared to other studies. On the

Table 6
Emissions of different biodiesels and blends in different engine types.

Ref.	Fuel	Engine	Remarks
[45]	Methyl ester neat rapeseed and its blend with fuel (5%, 10%, 20%, and 35%)	Unmodified direct injection diesel engine, with 4-cylinder and four strokes	Decreased CO, HC, and visible emissions but increased methyl ester compared to diesel.
[50]	Jatropha oil	Compression ignition engine	Hot exhaust recirculation for oxides of nitrogen control.
[52]	Methyl ester of polanga oil (20%, 40%, 60%, 80%, and 100%)	Single cylinder diesel engine	Compared to high-speed diesel, it showed lower exhaust emissions.
[96]	P0 (Premium Diesel Fuel), P10 (90% P0 + 10% palm biodiesel), P20, P30, P50, P75, and P100	Diesel engine	A low gross heat value for palm biodiesel caused a rise in BSFC.
[97]	Mixing peanut oil and sunflower oil with diesel fuel	Single-cylinder engine	When compared to 100% diesel fuel, it increased the carbon deposit number on the injector tip. Mass-based heating value is lower for vegetable oil compared to diesel fuel.
[98]	Rapeseed methyl ester	Diesel engine	Unburnt hydrocarbon emission was low.
[99]	Rapeseed methyl ester and palm oil	Direct injection diesel engine	Oxides of NOx (nitrogen emissions) and smoke were low.
[100]	Soybean methyl ester/blends (10%, 20%, 30%, and 40%)	Heavy-duty diesel engine	The 20/80 blend of biodiesel/diesel fuel was optimised based on the trade-off between NOx increase and PM decrease.
[101]	Neat rapeseed oil	Diesel engine	Durability characteristics and emissions for the exhaust.
[102]	Soybean ester	Direct injection diesel engine	Compared to diesel, emissions of NOx, CO, and smoke are lower slightly. Also, a 50% reduction in HC emissions.
[103]	Methyl ester of rapeseed oil and diesel blend	Single cylinder diesel engine	There is a considerable reduction in smoke, NOx, and CO emissions when the inlet air temperature increases beyond ambient temperature.
[104]	Blends of diesel with ethanol and tallowate methyl	522 kW direct-injection engine, 6-cylinder, and turbocharged	No change with CO_2 and HC emissions, but CO was reduced. Smoke and NOx emissions were like diesel. The diesel: tallowate methyl: ethanol (80:13:7) blend obtained lower emissions and the best performance.
[105]	Biodiesel-Ethanol-Diesel (BE-Diesel)	Cummins-4B diesel engine	In most operating conditions, BE-diesel had less THC, and a result of BE-diesel saw an increase of (2-14) % of NOx, but a significant decrease in PM emissions. A small amount of ethanol is observed in the exhaust when using BE-diesel.
[106]	Vegetable oil, blended (with frying oil, gas oil, and frying biodiesel oil)	Single cylinder, direct-injection engine	It was noted that nitrogen oxides decreased while maintaining the minimum value of BSFC when a biodiesel emulsion with 30% water was used.
[107]	Crude palm oil (CPO)	Diesel engine	There is no effect on the injection system or any benefit on the engine's performance when heating CPO to a high temperature. However, heating is necessary because the viscosity of CPO decreased with the preheating making it flow smoothly and avoiding fuel blockage.
[108]	Vegetable oil fuels and their methyl esters such as raw oils (sunflower, cottonseed, soybean vegetable oil fuels and their methyl esters (raw sunflower oil, raw cottonseed oil, raw soybean oil, and their methyl esters, refined corn oil distilled opium poppy oil and refined rapeseed oil)	Single-cylinder direct injection diesel engine	Vegetable oils and their esters are a promising alternative to the diesel engine. Vegetable oil fuel still suffers from problems such as atomisation, heavy particle emissions and flow due to its high viscosity, condensation in cold conditions and drying over time.
[109]	Waste cooking oil as neat biodiesel	Diesel engine	The HC emissions of the B100, B80, and diesel were more than the other fuel types, while the CO_2 was higher for B50 than other fuel types at all speeds of the engine except B20, which was the highest at 2000 rpm. B20 produced the lowest concentration of CO at all engine speeds.
[110]	Methyl ester CNG, DME, and diesel	Compression injection engine	Under conditions (fuel-rich, 800−1200 °C, at 24 atm).
[111]	Sunflower oil methyl ester/diesel blends	A stationary diesel engine	Emissions of NOx, CO, HC, and particulate matter were decreased.
[112]	Olive methyl ester	Direct injection diesel engine	Decreased smoke emission, decreased emissions by 59% CO, 8.6% CO_2, 32% NOx, and 57% sulphur.
[113]	Mustard seed oil	Tractor engine (4-cylinder direct ignition diesel engine)	Preheating oil caused decreased emissions, so preheating mustard oil for different speeds at all loads causes higher emissions for carbon monoxide.
[114]	Methyl ester of Karanja and 20%−80% blends with diesel	Direct injection diesel engine with a single cylinder	Lower emissions of CO, NO_x, and smoke.
[115]	Methyl ester of Pongamia	Compression injection engine with EGT (Exhaust Gas Temperature)	Effect on recirculation of exhaust gas and injection timing retardation.
[116]	Methyl ester of mahua oil (MOME)	Direct injection diesel engine (Kirloskar)	The use of MOME increased the BSFC by 20%, while the thermal efficiency decreased by 13% compared to diesel.
[117]	Methyl ester of tobacco seed oil	Diesel engine	SO_2 and CO emissions were reduced due to adding methyl ester of tobacco seed oil to diesel, while NO_x emissions increased slightly.
[118]	Rubber seed oil methyl ester	Internal combustion engine	It increased the concentration of biodiesel reduced exhaust gas emissions. Thus, for low blends, fuel consumption decreased, and BTE increased.
[119]	Cooking and unused vegetable oils methyl esters, biofuel and conventional fuel	Diesel engine	Analysis and measurement of smoke opacity
[120]	Jatropha oil	Diesel engine	Significant reduction of smoke and hydrocarbon emissions by increasing the injector-opening pressure and injector timing. It has reduced nitrogen oxide emissions.
[121]	Soy-oil methyl esters	Direct injection diesel engine	Decreased NO_x emissions based on the blends.
[95]	50% blend of papaya seed biodiesel	Single cylinder diesel engine	

Table 6 (*continued*)

Ref.	Fuel	Engine	Remarks
[122]	Waste cooking oil methyl ester (WCOME)	Katana KD4500 single-cylinder diesel engine	Decreased the emissions level for HC, CO, and smoke. But increased the NO_x emissions value. It reduced all the emissions values (CO, NO_x, and smoke)

Table 7
The description of biodiesel blending types.

Type of blends	Description	Notes	Examples
Biodiesel-diesel blends	Blends biodiesel which is produced from feedstock with diesel in different percentage volumes [8].	Increasing the ratio of biodiesel leads to a decrease in BTE while BP and BSFC increase [8].	Crude palm oil (B75) [123] Palm oil (PO75) Rapeseed oil (R 100) [124] Soybean oil (S100) [124] J. curcas (JB 10) [53] M. oleifera (MB10) [53] CTM30 (Calophyllum inophulhum oil) [125].
Mixed biodiesel-diesel blends	One way to enhance emission characteristics for diesel engines and the engine's performance is to reduce fossil fuel dependency and produce fuel from edible oils [23,24].	In general, these blends used not more than 30% of the total percentage volume of biodiesel [8].	1% Hazelnut-4% canola biodiesel-95% diesel blend [126]. 15% coconut-15% palm biodiesel-70% diesel blend [127]. 5% Palm-5% J. curcas biodiesel-90% diesel [128].
Biodiesel-diesel-bio-oil blends	Biodiesel-FPO (Fast Pyrolysis Oil)-diesel blends [129]. Biodiesel-bio-oils blends can be produced entirely from renewable sources, which maximised renewable chain benefits [130].	It has been shown that ignition delays cause poor combustion characteristics of internal combustion engines fuelled with bio-oil [8].	60:40 (biodiesel: fast pyrolysis oil) [118]. 10%:10%:80% (bio-oil: biodiesel: alcohol) [130] 40%:10%:50%) bio-oil: biodiesel: alcohol) [130].
Biodiesel-WTL blends	Using an MSW (Municipal Solid Waste) recycling technology to make WTL products from non-biodegradable materials [8].	PVC, (polyvinyl chloride), PP (polypropylene), PET (polypropylene-terephthalate), HDPE (high-density polyethylene), LDPE (low-density polyethylene, and PS (polystyrene) are non-biodegradable materials and petroleum-based. PS is widely used as it is easy to manufacture, has low cost, and is suitable for toys, packaging food, electronics, construction consumables, and medical consumables due to desirable mechanical properties [131].	P10 (10% of PPEH (pyrolysed polyethylene hydrocarbons)) with diesel) and P50 (50% PPEH blend with diesel) [132]. Blend 2% of methyl ester of soybean with P10 and P50 with 5% soybean methyl ester [132].
Blends of biodiesel-diesel-GTL	GTLs are synthetic liquid fuels derived from natural gas, which is a form of fossil fuel. The Fischer-Tropsch process is used to produce synthetic liquid fuel from natural gas [8].	Synthetic liquids contain negligible amounts of polyaromatics and sulphur and are primarily composed of long-chain paraffin molecules. GTLs have been shown in the studies to contain zero sulphur. The lower heating value was high, the kinematic viscosity is low, and the cetane number is high [133].	G + BD20 (80% GTL and 20% biodiesel, G + BD40 (60% GTL and 40% biodiesel) [133].
Biodiesel-alcohol blends	The alcohol fuel content reduced the combustion temperature, increasing the heat capacity of combustion products, and the oxidation process slows down [134].	Blending biodiesels with alcohols may compensate for the higher kinematic viscosities of biodiesel such as n-butanol, methanol, and ethanol [135,136].	Blends of palm biodiesel-surfactant-ethanol or palm biodiesel-methanol. The surfactant reduces the surface tension between the water and oil, their surfaces are activated, and superficial contact is maximised.
Biodiesel with nano-additives	Various applications used nano-additives such as nanomedicine, semiconductors, catalysts, sensors, batteries, and capacitors [137].	Make the fuels conductive, increase fuel viscosity, reduce knock, and enhance the burning rate by nano-additives. Conductivity indicates the ability to dissipate electrical charges, so this is a valuable	CeO_2-MWCTs (hybridised nano-cerium oxide-multi-walled carbon nanotubes) [138], and AL_2O_3 (aluminium oxide) [139] are examples of nano-additives. Methyl ester of jojoba use CeO_2-MWCTs

(*continued on next page*)

Table 7 (*continued*)

Type of blends	Description	Notes	Examples
		property. A low conductivity for fuel is highly undesirable to avoid accumulating static electrical charges, which may cause unexpected sparks [8].	additive [140]. Methyl ester of J. curcas emulsion fuel use CNTs (carbon nanotubes) additive [141].

other side, the rubber seed biodiesel [118] achieved the lowest BSFC value compared with diesel fuel. In contrast, the SOME [47] and Karanja biodiesel [51] gave the highest value compared to diesel fuel.

For future work, to have an equivalent comparison, the researchers need to apply different types of biofuel on the model of the same engine with different operating conditions to decide which biofuel will be the best for the environment and give the best engine performance. Also, to define which type of engine is helpful for biodiesel.

The parameters affecting the emissions can be summarised as follows:

i. Oxygen content, the biodiesel blend percentage, engine load, and engine type. When oxygen content is increased for the first parameter, the CO emissions and THC were reduced, but NOx was increased.
ii. CO emission will be reduced for the second parameter when the volume percentage blends are higher, but the NOx depends on the feedstock type. Most of the studies showed that NOx decreased. However, a rare few studies found that NOx decreased when they used mahua oil ethyl ester (MOEE), waste olive oil, jatropha biodiesel JB100, mahua methyl ester (MME) E20, and added MO and magnesium additive Mg to B60. Moreover, increasing the biodiesel blend content led to increased PM emissions.
iii. For the third parameter, most previous studies found that when the engine load is low, the CO emission is reduced, but NOx increased. However, a minority of researchers noticed that the NOx emission is reduced at whole loads of ranges. For example, one study using three types of waste oil biodiesel found that NOx was slightly lower at low load and increased at high load.

Table 8
Different biodiesel blends fuelled into different engine models.

Ref.	Biodiesel Blends	Engine
[11]	Soybean biodiesel and diesel	119, 205, and 276 kW engines
[22]	Pure ethyl esters and methyl esters soybean oil	Transient test cycle for 11.1 L engine
[33]	Soybean oil	TU, DI John Deere 4045T diesel engine with four cylinders and four strokes
[36]	Pure biodiesel and blends 75D, 50D, 25D	DI diesel engine with a single cylinder
[47]	JOME, SOME, and honge oil methyl ester HOME	WC, DI, and CI engine with a single-cylinder and four strokes
[51]	Blends based on methyl esters of Polanga and Karanja oil with diesel (KB20, KB50 and KB100) and blends for JB20, JB50, JB100, PoB20, PoB50 and PoB100	3-cylinder tractor engine
[114]	B100 (methyl ester of Karanja), and other blends (B20, B40, B60, and B80)	WC, DI diesel engine with a single-cylinder and four strokes
[142]	B20, B40, B60, B80, and B100 blends	WC diesel engine with six cylinders and four strokes
[143]	B20, B30, B50, B70, B80, B100 blends	DI diesel engine with 6 cylinders
[144]	Blends (B10, B20, B40, B60, B80, and B100)	NA and DI diesel engine with four strokes and three cylinders.
[145]	Methyl ester of cotton seed oil blends (CSOME): B5, B20, B50, B75, and B100	DI and AC diesel engine with a single cylinder
[146]	Soybean biodiesel and cooking oil	DI, TU engine
[147]	B10 blend (10% of chicken fat biodiesel with diesel)	DI diesel engine with a single cylinder
[148]	B20 and B100 blends	DI diesel engine with a single-cylinder, four strokes, and air-cooled
[149]	B10, B20, and B30 blends of Thumba oil biodiesel	WC and DI diesel engine with four cylinders and water-cooled
[150]	WCOE and WCOM (waste cooking oil ethyl ester and methyl ester) and their blends (WCOM30, WCOM70, WCOE30, and WCOE70)	2.2 L DI Nissan diesel engine with four cylinders and strokes, inter-cooled, and TU (turbocharged)
[151]	Pongamia biodiesel blends (B15 and B20)	Compression ignition engine
[152]	B10, B20, B30, B40, and B50 blends of waste cooking oil	Diesel engine with two cylinders and four strokes
[153]	Mahua biodiesel	Ricardo E6 engine with a single cylinder and four cylinders
[154]	HOME	Compression ignition diesel engine
[155]	Cooking oil methyl ester (CME100, CME20, and CME35)	IDI diesel engine with four strokes, cylinder
[156]	Pure biodiesel, diesel, and 20% biodiesel blend, pure biodiesel and their blends with 20% biodiesel	DI and IDI, TU and NA of heavy-duty engines
[157]	35% blend of soybean biodiesel with diesel	Tractor truck engines
[158]	B60 biodiesel blend	AC, DI diesel engine with a single cylinder and four strokes
[159]	B30, B50, and B100 blends	TU, 1.6L and HSDI diesel engine with four cylinders
[160]	Biodiesel-methanol blend BM5 (5% methanol), BM10 (10% methanol), and BM15 (15% methanol)	Direct injection diesel engine
[161]	Mahua methyl ester (MME) with 10% diesel and 10% ethanol (MME E10 D10), MME E10 (MME with 10% ethanol), MME E20, mahua methyl ester (MME)	CI, WC, NA engine with a single-cylinder and four strokes
[162]	C12:0 (lauric), C16:0 (palmitic), and C18:1 (oleic) methyl ester	6-cylinder engine
[163]	15% biodiesel, 5% bioethanol and 80% diesel (BD15E5) 20% biodiesel and 80% diesel (B20)	CRDI (Common Rail Direct) diesel engine
[164]	Blend pure palm oil biodiesel with 20% biodiesel and ultra-low-sulphur (ULS) diesel	2.84L, NA engine
[165]	Rapeseed biodiesel	4.2 kW engine with a single-cylinder
[166]	Diesel, using B100 and B20	DI and TU diesel engine
[144,153]	B10, B20, B40, B60, and B80 blends of fish oil biodiesel	AC, NA, DI diesel engine with three cylinders and four strokes

Table 9
Operation conditions for engines fuelled using biodiesel feedstocks.

Ref.	Biodiesel Blends	Engine	Operation hour
[119]	Blend of 20% rice bran methyl ester with mineral diesel (B20)	The in-cylinder engine has worn	100 h test
[167]	Waste olive oil (100%) and no. 2 diesel	WC, DI, 2.5L engine with three cylinders	Speed range between 1800 and 2100 rpm and 8–15 kW for 50 h
[168]	100% rapeseed oil and D2 diesel	WC, DI, 11L engine with six cylinders	110 h
[169]	100% Karanja, mahua oil	–	300 days
[170]	Palm oil (7.5%, 15%, and 100%)	WC, NA, IDI, 1.8L engine with 4 cylinders	100 h
[171,172]	20% linseed oil	1-cylinder, WC, portable	512 h

Table 10
Effect of the parameters on the emissions.

Parameters' influence on the emissions	Carbon Oxide CO	Nitrogen Oxide NO	Total unburnt hydrocarbon THC	Notes
Oxygen content	O higher, CO lower [53,173]	O higher, NO_x higher [174]	O higher, THC reduces [53]	The content of oxygen for biodiesel is high, improving combustion quality and increasing the temperature [8]. Soots is generated due to insufficient oxygen content; however, oxidation of soots occurred due to oxygen atoms in the biodiesel [174].
The volume percentage of biodiesel in the blends	(10:90) J. curcas with diesel, 14% CO reduces [53] (10:90) M. oleifera with diesel. 11% CO reduces [53]	Depend on the type of feedstock	Jatropha biodiesel JB10 blend reduces THC by 16% [53] MB10 blend reduces THC by 12%	Higher volume percentage blends will reduce CO [53]

iv. For the fourth parameter, a few reviewers mentioned that the engine model affected the emissions. For instance, using a common-rail engine reduced the NOx, while others found that NOx emissions increased in an LHR engine compared to the original engine. However, they were unanimous that the CO emissions were reduced in all conditions. The HC was also reduced during most experiments except for one study that showed an increase of 10%.

5.1. Recommendations

The study on biodiesel blends must be carried out to ensure optimisation in engine operation [5]. Enhancement of the performance for the engine and exhaust emission reduction was attained by improving the methods of various properties of biodiesels and modifying diesel engines. Moreover, each method has drawbacks and benefits depending on the government policy, natural potency, and environmental condition of the biodiesel implementation area. Further, exhaust emission reduction and optimising and choosing the improvement for the engine performance may impede the cost-effectiveness of each method in the future [8].

More research and study have to be carried out about long-term wear, injection duration and timing along with the tribological analysis for better combustion in diesel engines and properties of biodiesel fuel characteristics [9]. The feature of biodiesel injection is influential in the case the engine is not modified. Therefore, the study needs to be carried out to discover the relationship between injection timing and injection pressure on the engine power to obtain the optimal match while using biodiesel. The expectation of using biodiesel is to improve the engine durability compared to using diesel and lower inherent lubricity and soot formation. Despite that, there are not enough studies on endurance tests on the biodiesel engine to clarify the mechanism and reason for wears because the tests are time-consuming. Instead, further studies on the effect of injection pressure and injection time on the biodiesel emissions of NOx can be done.

More research about biodiesel properties and their impact on fuel and combustion systems for better illustration of increasing NOx is required. In the future, biodiesel production needs to be improved, and the properties and quality of biodiesel need to be promoted. New additives need to be developed to improve biodiesel consumption to favour economy, emissions (especially for NOx emissions), and power recovery. Redesign or adjustment of the engine and its system control for biodiesel has to be done. This is mainly to achieve more efficient combustion by EGR control and optimising injection and ignition, thus meeting biodiesel engine

Table 11
Particulate matter (PM) effect by using different fuel.

Ref.	Fuel	PM
[51]	Karanja (KB20, KB50, and KB100)	Decreased.
[51]	Jatropha biodiesel (JB20, JB50, and JB100)	Decreased.
[51]	Polanga biodiesel (PoB20, PoB50, and PoB100)	Decreased.
[159]	B30, B50 and B100 (4-cylinder, 1.6 L, turbo-diesel engine)	Decreased.
[175]	20% and 50% for 20% blends and 100% biodiesel	Decreased.
[111]	Pure biodiesel	Increased.
[145]	Cottonseed oil ester-diesel fuel (B5, B20 and B50)	PM emissions increased due to the increasing biodiesel content in blends, which caused higher viscosity and density, leading to deteriorating fuel atomisation. JP-5 reduced PM emissions.

Table 12
Biodiesel effect on NOx Emissions.

Ref.	Fuel	NOx	Engine	Condition	Remarks
[176]	B100	+15%	4-cylinder direct ignition diesel engine	High load	12% O, high gas temperature
[180]	Waste palm oil methyl ester (WPOME) COME	+22.13% + 6.48%	6-cylinder WC, NA, DI diesel engine	0.75 °CA 1.25°CA SOI (start of injection)	Advance SOI (start of injection) caused an increase in NOx
[181]	Vegetable oil methyl ester (VOME) fuels	Range +5.58% to 25.97 vs. petroleum diesel	Diesel engine	Higher yield	VOME has higher oxygen content and cetane number
[156]	Diesel, pure biodiesel, and 20% biodiesel	Small difference	TU, DI, NA, IDI, DI are heavy-duty engines	Full load	–
[157]	35% of soybean oil biodiesel, and diesel	Small difference	Heavy truck	Nine heavy trucks with two different types of engines	NOx increased at higher load, pressure, and temperature
[44]	MOEE	−12% vs. diesel	Four-stroke direct injection diesel engine	A whole range of loads	Spray characteristics, oxygen content, and adiabatic flame temperature affect NOx emissions
[112]	Waste olive oil	−20%	Diesel direct injection Perkins engine	8-mode test cycles and steady-state operations	Combustion temperature, oxygen concentration, and time determined NOx
[47]	HOME, JOME, SOME,	At 80% load vs. 1080 ppm diesel	Compression ignition engine	990 RPM 970 RPM 1000 RPM	NOx depends on peak temperature
[159]	B30 B50 B100	+20.6% +25.9% +44.8%	TU, HSDI, 1.6L diesel engine with 4 cylinders	Speed (1000, 1500, 2000, 2500, and 3000 rpm)	When the biodiesel is increased, NOx increased
[177]	B100 B50	+12.62% +1.84%	6076TF030 John Deere engine	Speed 1500 rpm	The highest NOx level was for B100 when the engine load was at 75 kW
[145]	CSOME (B5, B20, B50, and B75)	NOx emissions reduced with the increase in biodiesel content in the blends	Single-cylinder DI and AC diesel engine	Varying speeds (1250, 1500, 1750, 2000, 2250, and 2500) and full load	NOx emission decreases for all blends except B5
[52]	B20 B100	+2% −4%	Single cylinder diesel engine	Varying loads (0%, 20%, 40%, 60%, 80%, and 100%)	The compression ratio, reaction time and temperature are lower. The different geometry for the engine caused these differences
[51]	B20, B50, and B100 blends of Karanja, Palanga, and Jatropha oils	+ NOx (Karanja and Palanga oil and their blends) Jatropha biodiesel JB100 has lower NOx emission values than JB20	Three-cylinder tractor engine	Engine speeds (1200, 1800, and 2200 rpm)	NOx increases due to the presence of oxygen in biodiesel, which increases the temperature of the combustion gas
[45]	RME-diesel blends	Max NOx	NA, DI diesel engine with 4 cylinders and 4 strokes	Varying loads (0.1−0.8) and engine speeds	The maximum NOx emission is proportional to the oxygen ratio of biodiesel
[182]	SME 20% SME Diesel fuel No.1 (no oxygen)	+11.2% vs No. 2 diesel +0.6% vs No. 2 diesel, <6% for No. 1 diesel vs No. 2 diesel	Direct injection diesel engine	Contain 10.97% oxygen in mass	The injection directly affects NOx emissions, which in turn are affected by density
[22]	Pure ethyl esters, methyl esters	NOx increased	Transient test cycle 11.1L engine	Engine speed (500 −2100 rpm)	Due to the increase of unsaturated compounds and decrease of average carbon chain length
[183]	Rapeseed oil biodiesel	+21% with TU vs. NA	Four-stroke direct injection diesel engine	NA and TU conditions	Because the combustion temperatures are higher and more air entered the engine.
[179]	20% pure biodiesel blends with ULS diesel	Reduced NOx (using the common-rail engine)	Two high injection-pressure engines	Utilised EGR	The NOx emission was not affected when the cetane was increased
[155,184]	Cooking methyl ester [145], Honge oil methyl ester [174]	NOx increases in LHR engine vs. original engine	Compression ignition engine without (low heat rejection) LHR [174], indirect-injection diesel engine [155]	Injection timings (23°, 19°), optimum injection pressures (260 and 205 bar)	Because the combustion temperature is high
[185]	Commercial biodiesel	NOx decreases at load increases	DI, NA diesel outboard engine with 4-stroke and single-cylinder	Test cycle ISO C-3	Turbulence inside the cylinder increases due to the higher combustion and residence time of the species is lower in the high-temperature zones
[186]	Three different biodiesels from waste oil	1) NOx reduces slightly at low load 2) NOx increases at high load	Single-cylinder engine	At 2000 rpm	–
[187]	WFOME	NOx increases	Land Rover diesel engine	At max torque and max power speeds	Increased emissions depend on the higher volumetric efficiencies and exhaust temperature
[158]	Magnesium additive Mg and Mo added to B60 biodiesel	1) NOx increases at light load and max medium load 2) NOx reduced at increased speed	DI, AC diesel engine with a single-cylinder and 4-stroke	Mg at low speed No, at high speed	The combustion temperature increases due to the higher oxygen content and better combustion of biodiesel fuel, increasing NOx emissions
[117,188]	Tobacco seed oil methyl ester (TSOME), blends (0% −100%)	1) NOx increases at full load and 75% load 2) Gradually reduced at 50% load	Indirect injection diesel engine	Full and partial loads	Time, combustion temperature and oxygen concentration are the main factors affecting NOx emissions

Table 12 (*continued*)

Ref.	Fuel	NOx	Engine	Condition	Remarks
[189]	B20, B50, and RME vs ULS diesel	NOx higher reduction Lower increase of smoke	Lister-Petter TR1 diesel engine	Using EGR	Contributed to the high reduction of increasing CO_2 and H_2O and the RME combustion is retarded
[190]	B100 (soy biodiesel, canola biodiesel, and yellow grease biodiesel)	Slight difference in NOx	DI, NA diesel engine with a single-cylinder and 4 strokes	Using EGR	NOx decreased with increasing EGR
[171]	B10, B20, B50	Compared to baseline data for diesel without EGR, the NOx is lower	Direct injection diesel engine	Using EGR	NOx is reduced when using the EGR because it lowers the flame's temperature and concentrates the oxygen in the combustion chamber. In contrast, particulate emissions increase
[147]	Blend the synthetic Mg additive with 10% of chicken fat and diesel	+5%	DI diesel engine with a single cylinder	Speed from 1800 to 300 rpm Full load	NOx emission is negatively affected when the combustion time is longer, and the timing is early for SOC (start of combustion)
[191]	Blend B20 with 1% 4-NPAA additive	NOx emissions improved	Indirect injection diesel engine (Isuzu)	Speed range (1000 −4000 rpm)	The fuel combustion temperature is responsible for producing NOx emissions
[178]	BE5 (5% ethanol to biodiesel) B100 (soybean methyl ester)	+2.6% +12%	V-8 engine	Engine load (100%, 75%, 50%, and 10%) Speed (2700-800 rpm)	The benefit of using an ethanol blend is to reduce NOx emissions
[161]	Mahua methyl ester (MME) MME E20 MME E10	Low NOx emission Lowest	NA, WC, CI engine with 2 strokes and single-cylinder	Under cold climate Temperature range (0 −45 °C)	The highest NOx emission value is MME because it is an oxygenated fuel

Table 13
Biodiesel effect on CO emissions.

Ref.	Fuel	CO	Engine	Condition
[192]	Rapeseed oil	−50%	Daimler Chrysler diesel engine	Different fuels
[114]	B100 (methyl ester of Karanja), B80, B40, B20 blends with diesel	-(73−94) %	Direct injection diesel engine	Load (10%, 25%, 50%, 75%, 85%, and 100%)
[180]	WPOME COME	−86.89% −72.68%	Direct injection diesel engine	Constant speed (1500 rpm) and full load
[47]	JOME SOME HOME	0.155% 0.12% 0.145% All vs. 0.1125% diesel	WC, DI, and CI engine with a single-cylinder, 4 strokes	At 1500 rpm, 80% load
[183]	Rapeseed oil methyl ester	In NA, biodiesel and diesel have higher CO than TU	DI diesel engine	NA TU
[191]	B20X (1% of 4-NPAA additive into B20) B0 B20	0.1% 0.35% 0.2%	Indirect injection diesel engine (Isuzu)	Constant speed of 2250 rpm
[160]	BM5	−6%	Direct injection diesel engine	Different engine loads

needs. Studies have to be done about the biodiesel low-temperature performance due to its viscosity being higher than diesel due to the various primary zone equivalence ratio for biodiesel and diesel without any change in the fuel nozzle, affecting the emissions. Studies should also be carried out about the non-regulated biodiesel emissions to obtain a conclusive trend, especially carbon emissions. The instrumentation or methodology used for measurement to fulfil the expected requirements needs to be improved [11].

6. Conclusions

Even though different instruments or measurement techniques, different driving cycles or operation conditions, different reference diesel or biodiesel used, or various tested engines have been continuous developments for biodiesel emissions and engine performances. The most sustainable energy source and able to play a significant role in transportation needs is biodiesel. It can be produced domestically and from renewable sources. In the past ten years, biodiesel emissions and its performance for the engine are the aspects that the researchers have focused on. Biodiesel is the most popular renewable energy fuel which can replace diesel in diesel engines without modification.

Biodiesel is divided into non-edible and edible feedstocks, and they are used to produce biodiesel blends. Using blends is more cost-effective and can enhance fuel physicochemical properties. The types of research which have to be performed are engine performance, low-temperature, modern instrument, and modifying and optimising the engine-blending different types of biodiesel. These lead to benefits including enhanced exhaust emission characteristics, improved engine performance of diesel engines, dependency reduction on fossil fuels and edible oils for fuel production. The factors affecting the emissions are cetane number, lower sulphur, aromatic compounds, oxygen content (if oxygen increases, the emission will increase), engine type, and operation condition (cycle mode, load, etc.). The issues that faced biodiesel are limited land use and deforestation, food vs. fuel war, feedstock issue, sustainability, tough global competition, and oxidation. The biodiesel engine's durability aspect comprises fuel system problems, carbon deposit, and engine wears. Using additives improved

Table 14
Biodiesel effect on HC Emissions.

Ref.	Fuel	HC	Engine	Condition	Notes
[51]	Jatropha Karanja Polyanga JB20 JB50 JB100	20.73% 20.64% 6.75% −32.28% −18.19% −20.73%	Tractor engine with 2 cylinders	Cycle tests of 8 modes	An extra oxygen atom makes the biodiesel blends optimised during the combustion period, so the HC decreases when biodiesel is used.
[193]	Methyl ester of Jatropha oil	+10% vs. diesel	Direct injection diesel engine with a single cylinder	Crank angle range (340 −420°)	HC emissions are reduced when entering hydrogen
[47]	HOME, JOME and SOME	Increased HC vs. standard diesel	CI, WC, DI engine with 4 strokes and single-cylinder	1500 rpm	Contributed to low volatility and poor atomisation of biodiesels
[159]	B30 B50 B100	22.9% 17.7% 16.4%	Direct injection diesel engine	Engine speed (1000, 1500, 200, 2500, 3000 rpm), and MVEG-A cycle	Decreasing the heating value by pure biodiesel is related to the increased fuel consumption and higher HC emissions as it produces higher local fuel-to-air ratios. In terms of HC emissions, for biodiesel, the advantage is reduced by high catalyst efficiency.
[180]	WPOME COME	14.29% 9.52%	Direct injection diesel engine	1500 rpm and full load	SOI timing for biodiesel enhanced HC emissions
[194]	Vegetable oil methyl ester, rapeseed, soybean, and diesel	Decreased THC emissions	ECE R49 test cycle of heavy-duty engine	Test with and without catalyst	While the biodiesel percentage was increased, the decrease was sharp without the catalytic converter
[161]	MME MME E20 MME E10 MME E10 D10	−12.4% vs. diesel −9.15% −5.25%	CI, NA, WC engine with a single cylinder and 4 strokes	Cold climate Temperature range (0−45 °C)	Although the HC emission is slightly higher when MME is blended with ethanol, it remains lower than diesel.
[163]	15% biodiesel, 5% ethanol, 80% diesel (BD15E15)	THC lower vs. B20	CRDI (common rail direct injection) diesel engine	Load (24%, 50%, 75%, and 100%) Speed (2100 and 3800 rpm)	Increasing the ratio of biodiesel blend leads to a decrease in the percentage of HC emission
[160]	BM5, BM10	Lower compared to biodiesel	Direct injection diesel engine	Different engine loads	15% methanol (BM15) has lower HC emissions except at lower and highest engine load (0.08 and 0.70 MPA, respectively).

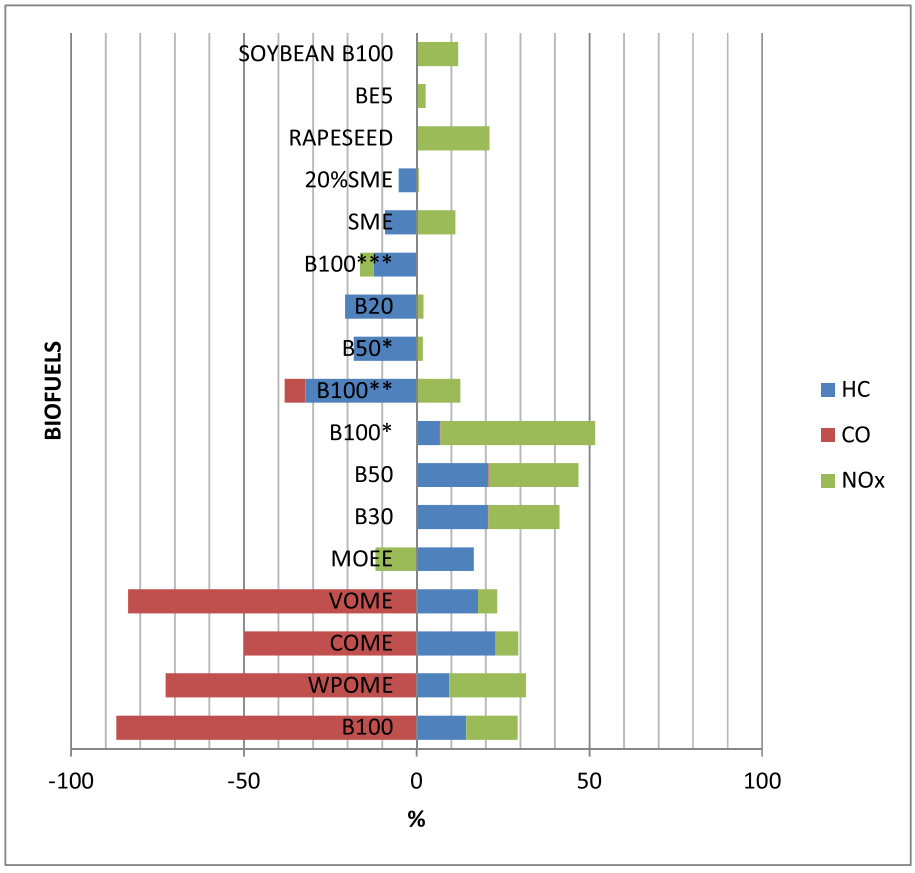


Fig. 4. Effect biofuel on engine emissions.

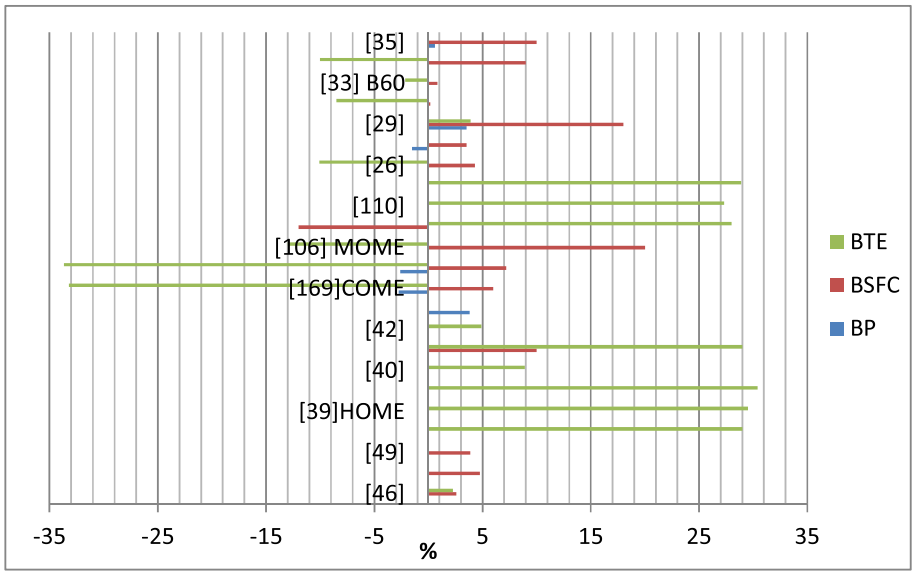


Fig. 5. Effect of biofuel on engine performance.

the biodiesel properties and combustion performance.

Compared to the short-term test, the long-term test showed engine failure due to higher carbon built up and lubricating oil contamination, so blending vegetable oils with diesel is essential to prevent this engine failure. The type of combustion chamber can also affect the results; it was found that the spherical combustion chamber gives better results. The biodiesel composition and its blends with diesel determine the range of the exhaust emissions and the engine performance parameters. The brake specific fuel consumption (BSFC) tends to be higher, while the brake thermal efficiency (BTE) and engine torque (ET) are lower when biodiesel-diesel blends are used.

Credit author statement

Yazan S. M. Altarazi: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Visualization. Abd Rahim Abu Talib: Resources, Writing – Review & Editing, Supervision, Project Administration, Funding Acquisition. Jianglong Yu: Supervision. Ezanee Gires: Supervision. Mohd Fahmi Abdul Ghafir: Supervision. John Lucas: Supervision. Talal Yusaf: Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Basha SA, Gopal KR, Jebaraj S. A review on biodiesel production, combustion, emissions and performance. Renew Sustain Energy Rev 2009;13(6–7):1628–34.
[2] Chiong MC, Chong CT, Ng J-H, Lam SS, Tran M-V, Chong WWF, et al. Liquid biofuels production and emissions performance in gas turbines: a review. Energy Convers Manag 2018;173:640–58.
[3] Enagi II, Al-Attab K, Zainal Z. Liquid biofuels utilization for gas turbines: a review. Renew Sustain Energy Rev 2018;90:43–55.
[4] Winter SR, Thropp JE, Rice S. What factors predict a consumer's support of environmental sustainability in aviation? A multi-model analysis. International Journal of Sustainable Aviation 2019;5(3):190–204.
[5] Silitonga A, Masjuki H, Mahlia T, Ong H, Chong W, Boosroh M. Overview properties of biodiesel diesel blends from edible and non-edible feedstock. Renew Sustain Energy Rev 2013;22:346–60.
[6] Altarazi YSM, Talib ARA, Gires E, Yu J, Lucas J, Yusaf T. Performance and exhaust emissions rate of small-scale turbojet engine running on dual biodiesel blends using Gasturb. Energy 2021:120971.
[7] Mekhilef S, Siga S, Saidur R. A review on palm oil biodiesel as a source of renewable fuel. Renew Sustain Energy Rev 2011;15(4):1937–49.
[8] Damanik N, Ong HC, Tong CW, Mahlia TMI, Silitonga AS. A review on the engine performance and exhaust emission characteristics of diesel engines fueled with biodiesel blends. Environ Sci Pollut Control Ser 2018:1–19.
[9] Ong H, Mahlia T, Masjuki H, Norhasyima R. Comparison of palm oil, Jatropha curcas and Calophyllum inophyllum for biodiesel: a review. Renew Sustain Energy Rev 2011;15(8):3501–15.
[10] Rains T, Winter SR, Rice S, Milner MN, Bledsaw Z, Anania EC. Biofuel and commercial aviation: will consumers pay more for it? International Journal of Sustainable Aviation 2017;3(3):217–32.
[11] Xue J, Grift TE, Hansen AC. Effect of biodiesel on engine performances and emissions. Renew Sustain Energy Rev 2011;15(2):1098–116.
[12] Dwivedi G, Jain S, Sharma M. Impact analysis of biodiesel on engine performance—a review. Renew Sustain Energy Rev 2011;15(9):4633–41.
[13] Khalil AE, Gupta AK. Toward ultra-low emission distributed combustion with fuel air dilution. Appl Energy 2015;148:187–95.
[14] Gupta K, Rehman A, Sarviya R. Bio-fuels for the gas turbine: a review. Renew Sustain Energy Rev 2010;14(9):2946–55.
[15] Du Toit M, Engelbrecht N, Oelofse SP, Bessarabov D. Performance evaluation and emissions reduction of a micro gas turbine via the co-combustion of H2/CH4/CO2 fuel blends. Sustainable Energy Technologies and Assessments 2020;39:100718.
[16] Cha S-H, Na S-I, Lee YH, Kim MS. Thermodynamic analysis of a gas turbine inlet air cooling and recovering system in gas turbine and CO2 combined cycle using cold energy from LNG terminal. Energy Convers Manag 2021;230:113802.
[17] Lee MC, Yoon Y. Development of a gas turbine fuel nozzle for DME and a design method thereof. Fuel 2012;102:823–30.
[18] Lee MC, Seo SB, Chung JH, Joo YJ, Ahn DH. Industrial gas turbine combustion performance test of DME to use as an alternative fuel for power generation. Fuel 2009;88(4):657–62.
[19] Lee MC, Seo SB, Chung JH, Joo YJ, Ahn DH. Combustion performance test of a new fuel DME to adapt to a gas turbine for power generation. Fuel 2008;87(10–11):2162–7.
[20] Ahmad A, Yasin NM, Derek C, Lim J. Microalgae as a sustainable energy source for biodiesel production: a review. Renew Sustain Energy Rev 2011;15(1):584–93.
[21] Azami MH, Savill M, Li Y-G. Comparison of aircraft engine performance and emission analysis using alternative fuels. International Journal of Sustainable Aviation 2017;3(1):43–63.
[22] Graboski M, Mccormick R, Alleman T, Herring A. The effect of biodiesel

composition on engine emissions from a DDC series 60 diesel engine. National Renewable Energy Laboratory 2003 (Report No: NREL/SR-510-31461).
[23] Damanik N, Ong HC, Chong W, Silitonga A. Biodiesel production from Calophyllum inophyllum– palm mixed oil. Energy Sources, Part A Recovery, Util Environ Eff 2017;39(12):1283–9.
[24] Atabani A, Mahlia T, Masjuki H, Badruddin IA, Yussof HW, Chong W, et al. A comparative evaluation of physical and chemical properties of biodiesel synthesized from edible and non-edible oils and study on the effect of biodiesel blending. Energy 2013;58:296–304.
[25] Unlu D, Hilmioglu ND. Potential fuel bioadditive ethyl levulinate production for the aviation fuel. International Journal of Sustainable Aviation 2016;2(4):338–47.
[26] Engler C. Compression ignition performance using sunflower seed oil. Winter Meeting ASAE; 1981. 1981.
[27] Ryan Iii T, Dodge L, Callahan T. The effects of vegetable oil properties on injection and combustion in two different diesel engines. Journal of the American Oil Chemists' Society 1984;61(10):1610–9.
[28] Bettis B, Peterson C, Auld D, Driscoll D, Peterson E. Fuel characteristics of vegetable oil from oilseed crops in the Pacific Northwest 1. Agron J 1982;74(2):335–9.
[29] Nwafor O, Rice G. Performance of rapeseed oil blends in a diesel engine. Appl Energy 1996;54(4):345–54.
[30] Strayer R, Blake J, Craig W. Canola and high erucic rapeseed oil as substitutes for diesel fuel: Preliminary tests. JAOCS (J Am Oil Chem Soc) 1983;60(8):1587–92.
[31] Sapuan S, Masjuki H, Azlan A. The use of palm oil as diesel fuel substitute. Proc IME J Power Energy 1996;210(1):47–53.
[32] Radu R, Mircea Z. The use of sunflower oil in Diesel engines. 1997. SAE Technical Paper 0148-7191.
[33] Raheman H, Ghadge S. Performance of compression ignition engine with mahua (Madhuca indica) biodiesel. Fuel 2007;86(16):2568–73.
[34] Niemi SA, Illikainen PE, Mäkinen ML, Laiho VO. Performance and exhaust emissions of a tractor engine using mustard seed oil as fuel. 1997. SAE Technical Paper 0148-7191.
[35] Mcdonnell KP, Ward SM, Mcnulty PB, Howard-Hildige R. Results of engine and vehicle testing of semirefined rapeseed oil. Transactions of the ASAE 2000;43(6):1309.
[36] Al-Widyan MI, Tashtoush G. Utilization of ethyl ester of waste vegetable oils as fuel in diesel engines. Fuel Process Technol 2002;76(2):91–103.
[37] Bari S, Yu C, Lim T. Performance deterioration and durability issues while running a diesel engine with crude palm oil. Proc Inst Mech Eng - Part D J Automob Eng 2002;216(9):785–92.
[38] Pramanik K. Properties and use of Jatropha curcas oil and diesel fuel blends in compression ignition engine. Renew Energy 2003;28(2):239–48.
[39] Ramadhas A, Jayaraj S, Muraleedharan C. Characterization and effect of using rubber seed oil as fuel in the compression ignition engines. Renew Energy 2005;30(5):795–803.
[40] Pradeep V, Sharma R. Evaluation of performance, emission and combustion parameters of a CI engine fuelled with bio-diesel from rubber seed oil and its blends. 2005. SAE Technical Paper 0148-7191.
[41] Pugazhvadivu M, Jeyachandran K. Investigations on the performance and exhaust emissions of a diesel engine using preheated waste frying oil as fuel. Renew Energy 2005;30(14):2189–202.
[42] Altıparmak D, Keskin A, Koca A, Gürü M. Alternative fuel properties of tall oil fatty acid methyl ester–diesel fuel blends. Bioresour Technol 2007;98(2):241–6.
[43] Kinney A, Clemente T. Modifying soybean oil for enhanced performance in biodiesel blends. Fuel Process Technol 2005;86(10):1137–47.
[44] Puhan S, Vedaraman N, Sankaranarayanan G, Ram BVB. Performance and emission study of Mahua oil (madhuca indica oil) ethyl ester in a 4-stroke natural aspirated direct injection diesel engine. Renew Energy 2005;30(8):1269–78.
[45] Labeckas G, Slavinskas S. The effect of rapeseed oil methyl ester on direct injection diesel engine performance and exhaust emissions. Energy Convers Manag 2006;47(13–14):1954–67.
[46] Jagtap SP, Pawar AN, Lahane S. Improving the usability of biodiesel blend in low heat rejection diesel engine through combustion, performance and emission analysis. Renew Energy 2020;155:628–44.
[47] Banapurmath N, Tewari P, Hosmath R. Performance and emission characteristics of a DI compression ignition engine operated on Honge, Jatropha and sesame oil methyl esters. Renew Energy 2008;33(9):1982–8.
[48] Jindal S, Nandwana B, Rathore N, Vashistha V. Experimental investigation of the effect of compression ratio and injection pressure in a direct injection diesel engine running on Jatropha methyl ester. Appl Therm Eng 2010;30(5):442–8.
[49] Kumar MS, Ramesh A, Nagalingam B. An experimental comparison of methods to use methanol and Jatropha oil in a compression ignition engine. Biomass Bioenergy 2003;25(3):309–18.
[50] Pradeep V, Sharma R. Use of HOT EGR for NOx control in a compression ignition engine fuelled with bio-diesel from Jatropha oil. Renew Energy 2007;32(7):1136–54.
[51] Sahoo P, Das L, Babu M, Arora P, Singh V, Kumar N, et al. Comparative evaluation of performance and emission characteristics of jatropha, karanja and polanga based biodiesel as fuel in a tractor engine. Fuel 2009;88(9):1698–707.
[52] Sahoo P, Das L, Babu M, Naik S. Biodiesel development from high acid value polanga seed oil and performance evaluation in a CI engine. Fuel 2007;86(3):448–54.
[53] Rahman MM, Hassan MH, Kalam MA, Atabani AE, Memon LA, Rahman SA. Performance and emission analysis of Jatropha curcas and Moringa oleifera methyl ester fuel blends in a multi-cylinder diesel engine. J Clean Prod 2014;65:304–10.
[54] Sharon H, Karuppasamy K, Kumar DS, Sundaresan A. A test on DI diesel engine fueled with methyl esters of used palm oil. Renew Energy 2012;47:160–6.
[55] Tan ES, Anwar M, Adnan R, Idris M. Biodiesel for gas turbine application—an atomization characteristics study. Advances in Internal Combustion Engines and Fuel Technologies 2013:213–43.
[56] Carareto ND, Kimura CY, Oliveira EC, Costa MC, Meirelles AJ. Flash points of mixtures containing ethyl esters or ethylic biodiesel and ethanol. Fuel 2012;96:319–26.
[57] Liaquat A, Masjuki H, Kalam M, Fazal M, Khan AF, Fayaz H, et al. Impact of palm biodiesel blend on injector deposit formation. Appl Energy 2013;111:882–93.
[58] Yakovlieva A, Boichenko S, Lejda K. Evaluation of jet engine operation parameters using conventional and alternative jet fuels. International Journal of Sustainable Aviation 2019;5(3):230–48.
[59] Sallevelt J, Gudde J, Pozarlik AK, Brem G. The impact of spray quality on the combustion of a viscous biofuel in a micro gas turbine. Appl Energy 2014;132:575–85.
[60] Prussi M, Chiaramonti D, Riccio G, Martelli F, Pari L. Straight vegetable oil use in Micro-Gas Turbines: system adaptation and testing. Appl Energy 2012;89(1):287–95.
[61] Chiaramonti D, Rizzo AM, Spadi A, Prussi M, Riccio G, Martelli F. Exhaust emissions from liquid fuel micro gas turbine fed with diesel oil, biodiesel and vegetable oil. Appl Energy 2013;101:349–56.
[62] Niguse Y, Agrawal A. Low-emission, liquid fuel combustion system for conventional and alternative fuels developed by the scaling analysis. J Eng Gas Turbines Power 2016;138(4):041502.
[63] Hashimoto N, Nishida H, Ozawa Y. Fundamental combustion characteristics of Jatropha oil as alternative fuel for gas turbines. Fuel 2014;126:194–201.
[64] Panchasara HV, Simmons BM, Agrawal AK, Spear SK, Daly DT. Combustion performance of biodiesel and diesel-vegetable oil blends in a simulated gas turbine burner. ASME Turbo Expo 2008:377–88. Power for Land, Sea, and Air, 2008.
[65] Kun-Balog A, Sztankó K. Reduction of pollutant emissions from a rapeseed oil fired micro gas turbine burner. Fuel Process Technol 2015;134:352–9.
[66] Józsa V, Kun-Balog A. Stability and emission analysis of crude rapeseed oil combustion. Fuel Process Technol 2017;156:204–10.
[67] Chiariello F, Allouis C, Reale F, Massoli P. Gaseous and particulate emissions of a micro gas turbine fuelled by straight vegetable oil–kerosene blends. Exp Therm Fluid Sci 2014;56:16–22.
[68] Chong CT, Hochgreb S. Spray combustion characteristics of palm biodiesel. Combust Sci Technol 2012;184(7–8):1093–107.
[69] Hashimoto N, Ozawa Y, Mori N, Yuri I, Hisamatsu T. Fundamental combustion characteristics of palm methyl ester (PME) as alternative fuel for gas turbines. Fuel 2008;87(15–16):3373–8.
[70] Erazo Jr JA, Parthasarathy R, Gollahalli S. Atomization and combustion of canola methyl ester biofuel spray. Fuel 2010;89(12):3735–41.
[71] Li L, Zhang X, Wu Z, Deng J, Huang C. Experimental study of biodiesel spray and combustion characteristics. 2006. SAE Technical Paper 0148-7191.
[72] Chong CT, Hochgreb S. Flame structure, spectroscopy and emissions quantification of rapeseed biodiesel under model gas turbine conditions. Appl Energy 2017;185:1383–92.
[73] Sequera D, Agrawal AK, Spear SK, Daly DT. Combustion performance of liquid bio-fuels in a swirl-stabilized burner. ASME Turbo Expo 2007:487–97. Power for Land, Sea, and Air, 2007.
[74] Bolszo C, Mcdonell V. Emissions optimization of a biodiesel fired gas turbine. Proc Combust Inst 2009;32(2):2949–56.
[75] Krishna C. Performance of the capstone C30 microturbine on biodiesel blends. Upton, NY (United States): Brookhaven National Lab. (BNL); 2007.
[76] Nascimento MA, Lora EES, Rendon MA. Performance and emission experimental evaluation and comparison of a regenerative gas microturbine using biodiesel from various sources as fuel. J Energy Resour Technol 2011;133(2):022204.
[77] Habib Z, Parthasarathy R, Gollahalli S. Performance and emission characteristics of biofuel in a small-scale gas turbine engine. Appl Energy 2010;87(5):1701–9.
[78] Ellis W, Lear W, Singh B, Srinivasan A, Crittenden J, Sherif S. Flameless combustion of biofuels in a semi-closed cycle gas turbine. In: 46th AIAA aerospace sciences meeting and exhibit; 2008. p. 1140.
[79] MolièRe M, Panarotto E, Aboujaib M, Bisseaud JM, Campbell A, Citeno J, et al. Gas turbines in alternative fuel applications: biodiesel field test. ASME Turbo Expo 2007:397–406. Power for Land, Sea, and Air, 2007.
[80] Pucher G, Allan W, Poitras P. Emissions from a gas turbine sector rig operated with synthetic aviation and biodiesel fuel. ASME Turbo Expo 2010:401–10. Power for Land, Sea, and Air, 2010.
[81] Corporan E, Reich R, Monroig O, Dewitt MJ, Larson V, Aulich T, et al. Impacts of biodiesel on pollutant emissions of a JP-8–fueled turbine engine. J Air Waste Manag Assoc 2005;55(7):940–9.

[82] Abu Talib AR, Gires E, Ahmad M. Performance evaluation of a small-scale turbojet engine running on palm oil biodiesel blends. Journal of Fuels 2014;2014.
[83] Timko MT, Herndon SC, De La Rosa Blanco E, Wood EC, Yu Z, Miake-Lye RC, et al. Combustion products of petroleum jet fuel, a Fischer–Tropsch synthetic fuel, and a biomass fatty acid methyl ester fuel for a gas turbine engine. Combust Sci Technol 2011;183(10):1039–68.
[84] Rehman A, Phalke DR, Pandey R. Alternative fuel for gas turbine: esterified jatropha oil–diesel blend. Renew Energy 2011;36(10):2635–40.
[85] Simmons BM, Agrawal AK. Flow blurring atomization for low-emission combustion of liquid biofuels. Combust Sci Technol 2012;184(5):660–75.
[86] Zhang Y, Van Gerpen JH. Combustion analysis of esters of soybean oil in a diesel engine. 1996. SAE Technical Paper 0148-7191.
[87] Radwan M, Dandoush S, Selim M, Kader A. Ignition delay period of jojoba diesel engine fuel. 1997. SAE Technical Paper 0148-7191.
[88] Ali Y, Hanna MA. In-cylinder pressure characteristics of a DI heavy duty diesel engine on biodiesel fuel. 1997. SAE Technical Paper 0148-7191.
[89] Yu C, Bari S, Ameen A. A comparison of combustion characteristics of waste cooking oil with diesel as fuel in a direct injection diesel engine. Proc Inst Mech Eng - Part D J Automob Eng 2002;216(3):237–43.
[90] Tashtoush G, Al-Widyan MI, Al-Shyoukh AO. Combustion performance and emissions of ethyl ester of a waste vegetable oil in a water-cooled furnace. Appl Therm Eng 2003;23(3):285–93.
[91] Sinha S, Agarwal AK. Combustion characteristics of rice bran oil derived biodiesel in a transportation diesel engine. In: ASME 2006 internal combustion engine division spring technical conference; 2006. p. 333–40.
[92] Usta N, Öztürk E, Can Ö, Conkur E, Nas S, Con A, et al. Combustion of biodiesel fuel produced from hazelnut soapstock/waste sunflower oil mixture in a diesel engine. Energy Convers Manag 2005;46(5):741–55.
[93] Lif A, Holmberg K. Water-in-diesel emulsions and related systems. Adv Colloid Interface Sci 2006;123:231–9.
[94] Szybist JP, Song J, Alam M, Boehman AL. Biodiesel combustion, emissions and emission control. Fuel Process Technol 2007;88(7):679–91.
[95] Devarajan Y, Beemkumar N, Ganesan S, Arunkumar T. An experimental study on the influence of an oxygenated additive in diesel engine fuelled with neat papaya seed biodiesel/diesel blends. Fuel 2020;268:117254.
[96] Lin Y-C, Lee W-J, Hou H-C. PAH emissions and energy efficiency of palm-biodiesel blends fueled on diesel generator. Atmos Environ 2006;40(21):3930–40.
[97] Barsic N, Humke A. Performance and emissions characteristics of a naturally aspirated diesel engine with vegetable oil fuels. SAE Trans 1981:1173–87.
[98] Nwafor O, Rice G. Performance of rapeseed methyl ester in diesel engine. Renew Energy 1995;6(3):335–42.
[99] Murayama T, Oh Y-T, Miyamoto N, Chikahisa T, Takagi N, Itow K. Low carbon flower buildup, low smoke, and efficient diesel operation with vegetable oils by conversion to mono-esters and blending with diesel oil or alcohols. SAE Trans 1984:292–302.
[100] Schumacher LG, Borgelt S, Fosseen D, Goetz W, Hires W. Heavy-duty engine exhaust emission tests using methyl ester soybean oil/diesel fuel blends. Bioresour Technol 1996;57(1):31–6.
[101] Hemmerlein N, Korte V, Richter H, Schröder G. Performance, exhaust emissions and durability of modern diesel engines running on rapeseed oil. SAE Trans 1991:400–15.
[102] Scholl KW, Sorenson SC. Combustion of soybean oil methyl ester in a direct injection diesel engine. SAE Trans 1993:1450–62.
[103] Desantes JM, Arregle J, Ruiz S, Delage A. Characterisation of the injection-combustion process in a DI diesel engine running with rape oil methyl ester. SAE Trans 1999:984–91.
[104] Ali Y, Hanna MA, Borg JE. Optimization of diesel, methyl tallowate and ethanol blend for reducing emissions from diesel engine. Bioresour Technol 1995;52(3):237–43.
[105] Shi X, Pang X, Mu Y, He H, Shuai S, Wang J, et al. Emission reduction potential of using ethanol–biodiesel–diesel fuel blend on a heavy-duty diesel engine. Atmos Environ 2006;40(14):2567–74.
[106] Yoshimoto Y, Onodera M, Tamaki H. Reduction of NOx, smoke, and BSFC in a diesel engine fueled by biodiesel emulsion with used frying oil. SAE Trans 1999:1913–20.
[107] Bari S, Lim T, Yu C. Effects of preheating of crude palm oil (CPO) on injection system, performance and emission of a diesel engine. Renew Energy 2002;27(3):339–51.
[108] Altın R, Cetinkaya S, Yücesu HS. The potential of using vegetable oil fuels as fuel for diesel engines. Energy Convers Manag 2001;42(5):529–38.
[109] Lin Y-F, Wu Y-PG, Chang C-T. Combustion characteristics of waste-oil produced biodiesel/diesel fuel blends. Fuel 2007;86(12–13):1772–80.
[110] Sidhu S, Graham J, Striebich R. Semi-volatile and particulate emissions from the combustion of alternative diesel fuels. Chemosphere 2001;42(5–7):681–90.
[111] Kalligeros S, Zannikos F, Stournas S, Lois E, Anastopoulos G, Teas C, et al. An investigation of using biodiesel/marine diesel blends on the performance of a stationary diesel engine. Biomass Bioenergy 2003;24(2):141–9.
[112] Dorado M, Ballesteros E, Arnal J, Gomez J, Lopez FJ. Exhaust emissions from a diesel engine fueled with transesterified waste olive oil. Fuel 2003;82(11):1311–5.
[113] Niemi SA, Murtonen TT, Lauren MJ, Laiho VO. Exhaust particulate emissions of a mustard seed oil driven tractor engine. SAE Trans 2002:335–46.
[114] Raheman H, Phadatare A. Diesel engine emissions and performance from blends of karanja methyl ester and diesel. Biomass Bioenergy 2004;27(4):393–7.
[115] Suryawanshi J, Deshpande N. Overview of EGR, injection timing and pressure on emissions and performance of CI engine with pongamia methyl ester. 2005. SAE Technical Paper 0148-7191.
[116] Puhan S, Vedaraman N, Ram BV, Sankarnarayanan G, Jeychandran K. Mahua oil (Madhuca Indica seed oil) methyl ester as biodiesel-preparation and emission characterstics. Biomass Bioenergy 2005;28(1):87–93.
[117] Usta N. An experimental study on performance and exhaust emissions of a diesel engine fuelled with tobacco seed oil methyl ester. Energy Convers Manag 2005;46(15–16):2373–86.
[118] Ramadhas A, Muraleedharan C, Jayaraj S. Performance and emission evaluation of a diesel engine fueled with methyl esters of rubber seed oil. Renew Energy 2005;30(12):1789–800.
[119] Armas O, Hernández JJ, Cárdenas MD. Reduction of diesel smoke opacity from vegetable oil methyl esters during transient operation. Fuel 2006;85(17–18):2427–38.
[120] Reddy JN, Ramesh A. Parametric studies for improving the performance of a Jatropha oil-fuelled compression ignition engine. Renew Energy 2006;31(12):1994–2016.
[121] Hess MA, Haas MJ, Foglia TA. Attempts to reduce NOx exhaust emissions by using reformulated biodiesel. Fuel Process Technol 2007;88(7):693–9.
[122] Bencheikh K, Atabani A, Shobana S, Mohammed M, Uğuz G, Arpa O, et al. Fuels properties, characterizations and engine and emission performance analyses of ternary waste cooking oil biodiesel–diesel–propanol blends. Sustainable Energy Technologies and Assessments 2019;35:321–34.
[123] Yusaf T, Yousif B, Elawad M. Crude palm oil fuel for diesel-engines: experimental and ANN simulation approaches. Energy 2011;36(8):4871–8.
[124] Celikten I, Koca A, Arslan MA. Comparison of performance and emissions of diesel fuel, rapeseed and soybean oil methyl esters injected at different pressures. Renew Energy 2010;35(4):814–20.
[125] Ong HC, Masjuki H, Mahlia T, Silitonga A, Chong W, Leong K. Optimization of biodiesel production and engine performance from high free fatty acid Calophyllum inophyllum oil in CI diesel engine. Energy Convers Manag 2014;81:30–40.
[126] Öztürk E. Performance, emissions, combustion and injection characteristics of a diesel engine fuelled with canola oil–hazelnut soapstock biodiesel mixture. Fuel Process Technol 2015;129:183–91.
[127] Habibullah M, Masjuki HH, Kalam M, Fattah IR, Ashraful A, Mobarak H. Biodiesel production and performance evaluation of coconut, palm and their combined blend with diesel in a single-cylinder diesel engine. Energy Convers Manag 2014;87:250–7.
[128] Sanjid A, Masjuki H, Kalam M, Rahman SA, Abedin M, Palash S. Production of palm and jatropha based biodiesel and investigation of palm-jatropha combined blend properties, performance, exhaust emission and noise in an unmodified diesel engine. J Clean Prod 2014;65:295–303.
[129] Laesecke J, Ellis N, Kirchen P. Production, analysis and combustion characterization of biomass fast pyrolysis oil–Biodiesel blends for use in diesel engines. Fuel 2017;199:346–57.
[130] Alcalá A, Bridgwater A. Upgrading fast pyrolysis liquids: blends of biodiesel and pyrolysis oil. Fuel 2013;109:417–26.
[131] Miandad R, Barakat M, Aburiazaiza AS, Rehan M, Ismail I, Nizami A. Effect of plastic waste types on pyrolysis liquid oil. Int Biodeterior Biodegrad 2017;119:239–52.
[132] Sharma BK, Moser BR, Vermillion KE, Doll KM, Rajagopalan N. Production, characterization and fuel properties of alternative diesel fuel from pyrolysis of waste plastic grocery bags. Fuel Process Technol 2014;122:79–90.
[133] Moon G, Lee Y, Choi K, Jeong D. Emission characteristics of diesel, gas to liquid, and biodiesel-blended fuels in a diesel engine for passenger cars. Fuel 2010;89(12):3840–6.
[134] Dharma S, Ong HC, Masjuki H, Sebayang A, Silitonga A. An overview of engine durability and compatibility using biodiesel–bioethanol–diesel blends in compression-ignition engines. Energy Convers Manag 2016;128:66–81.
[135] Ali O, Yusaf T, Mamat R, Abdullah N, Abdullah A. Influence of chemical blends on palm oil methyl esters' cold flow properties and fuel characteristics. Energies 2014;7(7):4364–80.
[136] Lapuerta M, Rodríguez-Fernández J, Fernández-Rodríguez D, Patiño-Camino R. Modeling viscosity of butanol and ethanol blends with diesel and biodiesel fuels. Fuel 2017;199:332–8.
[137] Piriyawong V, Thongpool V, Asanithi P, Limsuwan P. Preparation and characterization of alumina nanoparticles in deionized water using laser ablation technique. J Nanomater 2012;2012:2.
[138] Mirzajanzadeh M, Tabatabaei M, Ardjmand M, Rashidi A, Ghobadian B, Barkhi M, et al. A novel soluble nano-catalysts in diesel–biodiesel fuel blends to improve diesel engines performance and reduce exhaust emissions. Fuel 2015;139:374–82.
[139] Hosseini SH, Taghizadeh-Alisaraei A, Ghobadian B, Abbaszadeh-Mayvan A. Effect of added alumina as nano-catalyst to diesel-biodiesel blends on performance and emission characteristics of CI engine. Energy 2017;124:543–52.
[140] El-Seesy AI, Abdel-Rahman AK, Bady M, Ookawara S. Performance, combustion, and emission characteristics of a diesel engine fueled by biodiesel-diesel mixtures with multi-walled carbon nanotubes additives. Energy Convers Manag 2017;135:373–93.

[141] Basha JS, Anand R. Performance, emission and combustion characteristics of a diesel engine using Carbon Nanotubes blended Jatropha Methyl Ester Emulsions. Alexandria Engineering Journal 2014;53(2):259–73.
[142] Reyes JF, Sepulveda M. PM-10 emissions and power of a diesel engine fueled with crude and refined biodiesel from salmon oil. Fuel 2006;85(12–13):1714–9.
[143] Carraretto C, Macor A, Mirandola A, Stoppato A, Tonon S. Biodiesel as alternative fuel: experimental analysis and energetic evaluations. Energy 2004;29(12–15):2195–211.
[144] Godiganur S, Murthy CS, Reddy RP. Performance and emission characteristics of a Kirloskar HA394 diesel engine operated on fish oil methyl esters. Renew Energy 2010;35(2):355–9.
[145] Aydin H, Bayindir H. Performance and emission analysis of cottonseed oil methyl ester in a diesel engine. Renew Energy 2010;35(3):588–92.
[146] Canakci M, Van Gerpen JH. Comparison of engine performance and emissions for petroleum diesel fuel, yellow grease biodiesel, and soybean oil biodiesel. Transactions of the ASAE 2003;46(4):937.
[147] Gürü M, Koca A, Can Ö, Çınar C, Şahin F. Biodiesel production from waste chicken fat based sources and evaluation with Mg based additive in a diesel engine. Renew Energy 2010;35(3):637–43.
[148] Gumus M, Kasifoglu S. Performance and emission evaluation of a compression ignition engine using a biodiesel (apricot seed kernel oil methyl ester) and its blends with diesel fuel. Biomass Bioenergy 2010;34(1):134–9.
[149] Pal A, Verma A, Kachhwaha S, Maji S. Biodiesel production through hydrodynamic cavitation and performance testing. Renew Energy 2010;35(3):619–24.
[150] Lapuerta M, Herreros JM, Lyons LL, García-Contreras R, Briceño Y. Effect of the alcohol type used in the production of waste cooking oil biodiesel on diesel performance and emissions. Fuel 2008;87(15–16):3161–9.
[151] Mahanta P, Mishra S, Kushwah Y. An experimental study of Pongamia pinnata L. oil as a diesel substitute. Proc IME J Power Energy 2006;220(7):803–8.
[152] Ghobadian B, Rahimi H, Nikbakht A, Najafi G, Yusaf T. Diesel engine performance and exhaust emission analysis using waste cooking biodiesel fuel with an artificial neural network. Renew Energy 2009;34(4):976–82.
[153] Sharma D, Soni S, Mathur J. Emission reduction in a direct injection diesel engine fueled by neem-diesel blend. Energy Sources, Part A 2009;31(6):500–8.
[154] Banapurmath N, Tewari P, Hosmath R. Effect of biodiesel derived from Honge oil and its blends with diesel when directly injected at different injection pressures and injection timings in single-cylinder water-cooled compression ignition engine. Proc IME J Power Energy 2009;223(1):31–40.
[155] Hazar H. Effects of biodiesel on a low heat loss diesel engine. Renew Energy 2009;34(6):1533–7.
[156] Durbin TD, Norbeck JM. Effects of biodiesel blends and Arco EC-diesel on emissions from light heavy-duty diesel vehicles. Environmental Science & Technology 2002;36(8):1686–91.
[157] Wang W-G, Lyons DW, Clark NN, Gautam M, Norton P. Emissions from nine heavy trucks fueled by diesel and biodiesel blend without engine modification. Environmental Science & Technology 2000;34(6):933–9.
[158] Keskin A, Gürü M, Altıparmak D. Influence of tall oil biodiesel with Mg and Mo based fuel additives on diesel engine performance and emission. Bioresour Technol 2008;99(14):6434–8.
[159] Lujan J, Bermúdez V, Tormos B, Pla B. Comparative analysis of a DI diesel engine fuelled with biodiesel blends during the European MVEG-A cycle: performance and emissions (II). Biomass Bioenergy 2009;33(6–7):948–56.
[160] Cheung C, Zhu L, Huang Z. Regulated and unregulated emissions from a diesel engine fueled with biodiesel and biodiesel blended with methanol. Atmos Environ 2009;43(32):4865–72.
[161] Bhale PV, Deshpande NV, Thombre SB. Improving the low temperature properties of biodiesel fuel. Renew Energy 2009;34(3):794–800.
[162] Knothe G, Sharp CA, Ryan TW. Exhaust emissions of biodiesel, petrodiesel, neat methyl esters, and alkanes in a new technology engine. Energy & Fuels 2006;20(1):403–8.
[163] Kim H, Choi B. The effect of biodiesel and bioethanol blended diesel fuel on nanoparticles and exhaust emissions from CRDI diesel engine. Renew Energy 2010;35(1):157–63.
[164] Choi S-H, Oh Y. The emission effects by the use of biodiesel fuel. Int J Mod Phys B 2006;20(25n27):4481–6.
[165] Munack A, Schröder O, Krahl J, Bünger J. Comparison of relevant gas emissions from biodiesel and fossil diesel fuel. 001 agricultural engineering international: the CIGR journal of scientific research and development 2001;1. III: Manuscript EE.
[166] He C, Ge Y, Tan J, You K, Han X, Wang J. Characteristics of polycyclic aromatic hydrocarbons emissions of diesel engine fueled with biodiesel and diesel. Fuel 2010;89(8):2040–6.
[167] Dorado M, Ballesteros E, Arnal J, Gomez J, López Giménez F. Testing waste olive oil methyl ester as a fuel in a diesel engine. Energy & Fuels 2003;17(6):1560–5.
[168] Pehan S, Jerman MS, Kegl M, Kegl B. Biodiesel influence on tribology characteristics of a diesel engine. Fuel 2009;88(6):970–9.
[169] Kaul S, Saxena R, Kumar A, Negi M, Bhatnagar A, Goyal H, et al. Corrosion behavior of biodiesel from seed oils of Indian origin on diesel engine parts. Fuel Process Technol 2007;88(3):303–7.
[170] Kalam M, Masjuki H. Biodiesel from palmoil—an analysis of its properties and potential. Biomass Bioenergy 2002;23(6):471–9.
[171] Agarwal D, Sinha S, Agarwal AK. Experimental investigation of control of NOx emissions in biodiesel-fueled compression ignition engine. Renew Energy 2006;31(14):2356–69.
[172] Agarwal AK, Bijwe J, Das L. Wear assessment in a biodiesel fueled compression ignition engine. Transactions-American Society of Mechanical Engineers Journal of Engineering for Gas Turbines and Power 2003;125:820–6.
[173] Nabi MN, Zare A, Hossain FM, Ristovski ZD, Brown RJ. Reductions in diesel emissions including PM and PN emissions with diesel-biodiesel blends. J Clean Prod 2017;166:860–8.
[174] Jiaqiang E, Liu T, Yang W, Deng Y, Gong J. A skeletal mechanism modeling on soot emission characteristics for biodiesel surrogates with varying fatty acid methyl esters proportion. Appl Energy 2016;181:322–31.
[175] Haas MJ, Scott KM, Alleman TL, Mccormick RL. Engine performance of biodiesel fuel prepared from soybean soapstock: a high quality renewable fuel produced from a waste feedstock. Energy & Fuels 2001;15(5):1207–12.
[176] Nabi MN, Hoque SN, Akhter MS. Karanja (Pongamia Pinnata) biodiesel production in Bangladesh, characterization of karanja biodiesel and its effect on diesel emissions. Fuel Process Technol 2009;90(9):1080–6.
[177] Lertsathapornsuk V, Pairintra R, Aryusuk K, Krisnangkura K. Microwave assisted in continuous biodiesel production from waste frying palm oil and its performance in a 100 kW diesel generator. Fuel Process Technol 2008;89(12):1330–6.
[178] Hansen AC, Gratton M, Yuan W. Diesel engine performance and NOx emissions from oxygenated biofuels and blends with diesel fuel. Transactions of the ASABE 2006;49(3):589–95.
[179] McCormick RL, Tennant CJ, Hayes RR, Black S, Ireland J, Mcdaniel T, et al. Regulated emissions from biodiesel tested in heavy-duty engines meeting 2004 emission standards. 2005. SAE Technical Paper 0148-7191.
[180] Ozsezen AN, Canakci M, Turkcan A, Sayin C. Performance and combustion characteristics of a DI diesel engine fueled with waste palm oil and canola oil methyl esters. Fuel 2009;88(4):629–36.
[181] Lin B-F, Huang J-H, Huang D-Y. Experimental study of the effects of vegetable oil methyl ester on DI diesel engine performance characteristics and pollutant emissions. Fuel 2009;88(9):1779–85.
[182] Alam M, Song J, Acharya R, Boehman A, Miller K. Combustion and emissions performance of low sulfur, ultra low sulfur and biodiesel blends in a DI diesel engine. SAE Trans 2004:1986–97.
[183] Karabektas M. The effects of turbocharger on the performance and exhaust emissions of a diesel engine fuelled with biodiesel. Renew Energy 2009;34(4):989–93.
[184] Banapurmath N, Tewari P. Performance of a low heat rejection engine fuelled with low volatile Honge oil and its methyl ester (HOME). Proc IME J Power Energy 2008;222(3):323–30.
[185] Murillo S, Miguez J, Porteiro J, Granada E, Moran J. Performance and exhaust emissions in the use of biodiesel in outboard diesel engines. Fuel 2007;86(12–13):1765–71.
[186] Hamasaki K, Kinoshita E, Tajima S, Takasaki K, Morita D. Combustion characteristics of diesel engines with waste vegetable oil methyl ester 5th Int. In: Symp on diagnostics and modelling of combustion in internal combustion engines; 2001.
[187] Utlu Z, Koçak MS. The effect of biodiesel fuel obtained from waste frying oil on direct injection diesel engine performance and exhaust emissions. Renew Energy 2008;33(8):1936–41.
[188] Usta N. Use of tobacco seed oil methyl ester in a turbocharged indirect injection diesel engine. Biomass Bioenergy 2005;28(1):77–86.
[189] Tsolakis A, Megaritis A, Wyszynski M, Theinnoi K. Engine performance and emissions of a diesel engine operating on diesel-RME (rapeseed methyl ester) blends with EGR (exhaust gas recirculation). Energy 2007;32(11):2072–80.
[190] Zheng M, Mulenga MC, Reader GT, Wang M, Ting DS, Tjong J. Biodiesel engine performance and emissions in low temperature combustion. Fuel 2008;87(6):714–22.
[191] Kalam M, Masjuki H. Testing palm biodiesel and NPAA additives to control NOx and CO while improving efficiency in diesel engines. Biomass Bioenergy 2008;32(12):1116–22.
[192] Krahl J, Munack A, Schröder O, Stein H, Bünger J. Influence of biodiesel and different designed diesel fuels on the exhaust gas emissions and health effects. SAE Trans 2003:2447–55.
[193] Kumar MS, Ramesh A, Nagalingam B. A comparison of the different methods of using Jatropha oil as fuel in a compression ignition engine. J Eng Gas Turbines Power 2010;132(3):032801.
[194] Aakko P, Nylund N-O, Westerholm M, Marjamäki M, Moisio M, Hillamo R, et al. Emissions from heavy-duty engine with and without aftertreatment using selected biofuels. In: FISITA 2002 world automotive congress proceedings; 2002. F02E195.